

Modelling Financial Markets

Chapter 9: Realised Volatility



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From Intraday Prices to Variance Forecasts (1/2)

What we will explain

- Part I, **measurement and diagnosis**: realised variance and its precision, microstructure noise, jumps
- Part II, **modelling and forecasting**: stylised facts, the HAR model, forecast evaluation against GARCH
  - HAR: heterogeneous autoregressive
  - GARCH: generalised autoregressive conditional heteroskedasticity (Chapter 5)
- Extensions: rough volatility, a machine-learning case study, research assisted by AI
  - AI: artificial intelligence
- Appendix: asymptotic theory, noise- and jump-robust estimators, model extensions
- Detailed examples and advanced topics are in Further Topics, after the Conclusions
  - the footer button leads there; the Back button returns to the starting slide

From Intraday Prices to Variance Forecasts (2/2)

What you should be able to do

- compute daily realised variance, its standard error and a confidence interval from intraday prices
- diagnose microstructure noise with a signature plot
- test for jumps with false discovery rate (FDR) control
- estimate log-HAR
- compare variance forecasts
  - with the quasi-likelihood loss (QLIKE)
  - with Diebold–Mariano, Mincer–Zarnowitz and the model confidence set

Volatility Is Latent: Can We Measure It? (1/2)



Torben G. Andersen, 2012

Photo: Soralum (2012); CC BY-SA 4.0; Wikimedia Commons

- Volatility drives risk, option prices and portfolio weights, but it is never observed
  - risk: Chapters 7–8; option prices: Chapter 12; portfolio weights: Chapter 4
  - One daily return is a single draw from a distribution whose dispersion we want to estimate
  - GARCH (Chapter 5) **infers** today's variance from past daily returns through a model
- Idea: the sum of squared intraday price changes **measures** the day's variance almost without a model Andersen & Bollerslev (1998)
  - inside one day there are hundreds of price changes
  - Early idea: Merton (1980) showed that variance, unlike the mean, is estimated better with more frequent data

Volatility Is Latent: Can We Measure It? (2/2)

- Two different objects
  - realised variance: an **ex-post measurement** of day t, known after the close
  - GARCH variance: a **conditional forecast** of day t made with the information of day $t-1$
- Measure first, then model
  - Andersen, Bollerslev, Diebold & Labys (2001), Andersen, Bollerslev, Diebold & Ebens (2001) study the distribution of realised volatility
  - Andersen, Bollerslev, Diebold & Labys (2003) forecast it directly
  - textbook treatment: Franke, Härdle & Hafner (2019)

Reading Intraday Data

- SPY: an Exchange-Traded Fund (ETF) tracking the Standard & Poor's 500 stock index
  - ETF: a fund whose units trade on an exchange
- UTC means Coordinated Universal Time
  - Bitcoin daily intervals run midnight to midnight in that time zone
- VIX is the Volatility Index of the Chicago Board Options Exchange
  - an option-price measure of expected volatility
- All series end on 18 September 2026
- Returns $r_i = 100(\ln P_i - \ln P_{i-1})$ in %, P_i: the price at the end of bar i
- Charts and times quoted in this chapter mark the **end** of each 5-minute interval
  - 09:35 is the first SPY return, 09:30–09:35

▸ Appendix: data series table

A Model for Intraday Prices (1/2)

- **Question**: how do we turn a day's 5-minute returns into a measurement of that day's variance?
- Notation: day t has M intervals of length $\Delta = 1/M$, returns $r_{t,i}$, daily return $r_t = \sum_i r_{t,i}$
  - Variances in $\%^2$
  - annualised volatility $\sqrt{252 \times \text{variance}}$ (SPY), $\sqrt{365 \times \text{variance}}$ (Bitcoin)
  - the SPY overnight return (close to next open) is outside the bars
- Log price in continuous time: $dp(s) = \mu(s)\,ds + \sigma(s)\,dW(s)$, with W a Brownian motion (Chapter 11)
  - $\sigma(s)$ is the **spot volatility**: it changes within the day and from day to day

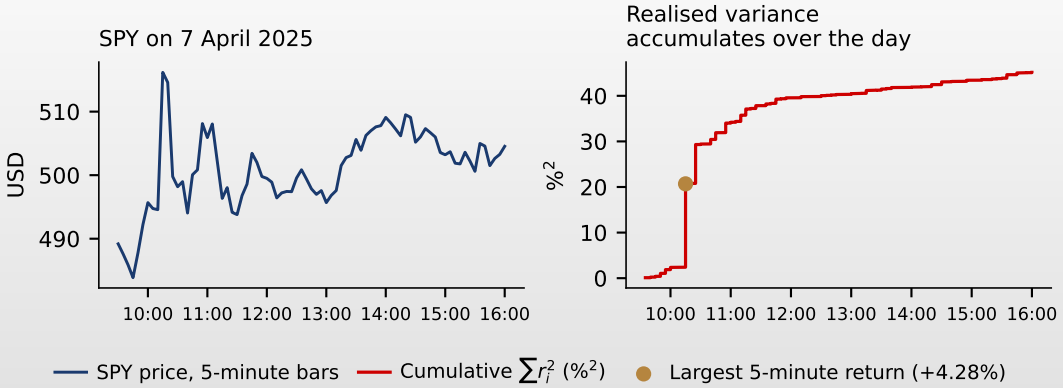
A Model for Intraday Prices (2/2)

- **Integrated variance** of day t: the quantity we want to measure
  - $\mathrm{IV}_t = \int_{t-1}^{t} \sigma^2(s)\, ds$: the spot variance accumulated over the day
  - in this chapter IV always means integrated variance, never implied volatility
- Given the path of σ, the daily return is Normal: $r_t \mid \sigma \sim N(\int \mu, \mathrm{IV}_t)$
  - assumption: σ is independent of W (no leverage)
  - assumption: the drift is zero (or fixed by the path of σ)
  - With leverage (Appendix: Itô semimartingales) this conditional normality fails, but the limit theory of realised variance (RV, next slide) still holds

Definition: Realised Variance

- **Realised variance**: the sum of squared intraday returns
  - $\mathrm{RV}_t = \sum_{i=1}^{M} r_{t,i}^2$, realised volatility $= \sqrt{\mathrm{RV}_t}$
- Why squares: RV_t is a Riemann sum for $\int_{t-1}^{t} \sigma^2(s)\, ds$
  - over a short interval $\mathbb{E}[r_{t,i}^2 \mid \sigma] \approx \sigma^2(s_i)\,\Delta$
- **Quadratic variation**: $[p]_t = \operatorname{plim}_{M\to\infty} \sum_{i=1}^{M} r_{t,i}^2 = \mathrm{IV}_t$ for a price without jumps Barndorff-Nielsen & Shephard (2002)
  - The drift adds terms of order Δ per day, which vanish as $M \to \infty$; for SPY the mean 5-minute return is below 1% of its standard deviation
- No parameters to estimate: RV_t is a **model-free** measurement of one day's variance
- The squared daily return r_t^2 is the special case $M = 1$
  - unbiased (with zero drift) but extremely noisy

One Day, 78 Returns: 7 April 2025 (1/2)



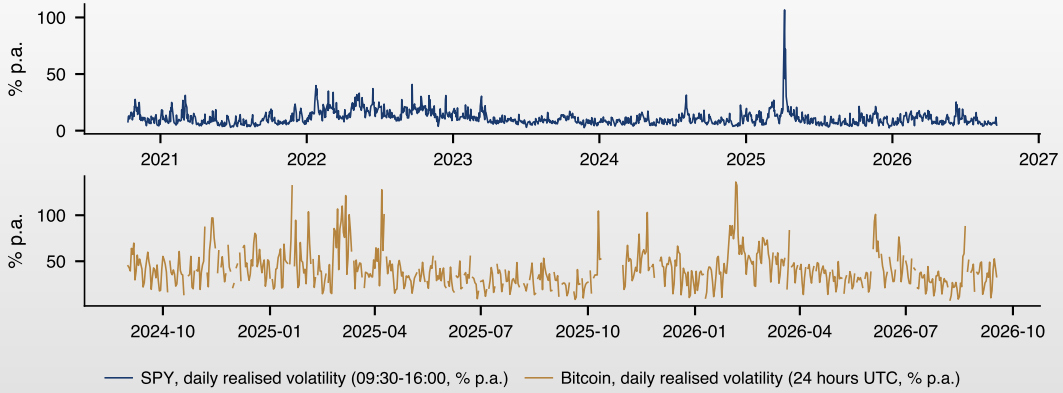
MFM_ch9_intraday_day

One Day, 78 Returns: 7 April 2025 (2/2)

- Left: SPY on 5-minute bars on the most volatile day of the sample
- Right: running sum of squared returns
- SPY opened with an overnight log return of -3.27%, after tariff announcements
  - 78 five-minute returns followed
  - $\mathrm{RV} = 45.1\%^2$: annualised realised volatility $\sqrt{252 \times 45.1} = 107\%$
- One 5-minute return, of $+4.28\%$, contributed 41% of the day's realised variance
  - interval ending 10:15
- Open-to-close the day ended $+3.09\%$
  - the daily return hides the intraday swings that RV records

MFM_ch9_intraday_day

Daily Realised Volatility: SPY and Bitcoin (1/2)



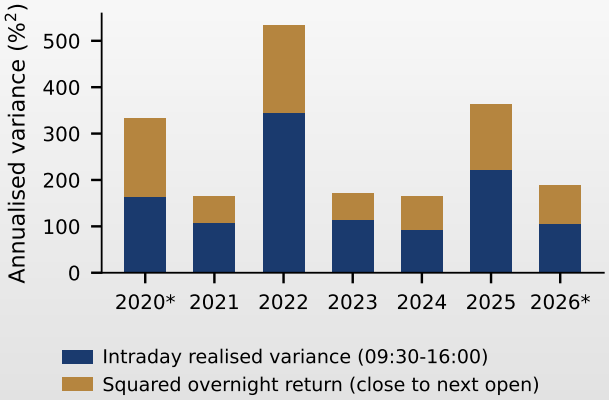
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Daily Realised Volatility: SPY and Bitcoin (2/2)

- Annualised, from 5-minute returns
  - SPY: regular US session
  - Bitcoin: 24 hours
  - the gap in October 2025 is missing data
- SPY: volatility from mean RV $\sqrt{252\,\overline{\mathrm{RV}}} = 12.9\%$ p.a.
  - median day 9.7%, range 2.4% to 106.7%
  - Volatility computed from mean variance exceeds the median: a few very volatile days raise the root-mean-square of daily volatilities
- Bitcoin: $\sqrt{365\,\overline{\mathrm{RV}}} = 45.1\%$ p.a.
  - median 37.8%
  - maximum 136.1% on 2026-02-05
- Both series show **clustering**, as with GARCH (Chapter 5)
  - but every point is a measurement, not a model output

MFM_ch9_rv_series

Where Is the Risk: Day or Night? (1/2)



MFM_ch9_overnight

Where Is the Risk: Day or Night? (2/2)

- SPY, mean daily variance by year, annualised
  - intraday realised variance plus squared overnight return
  - 2020 and 2026 are partial years
- Overnight returns carry 38% of SPY's daily variance, although the market is closed for 17.5 hours
  - News (earnings, macro data, Asian and European sessions) accumulates and is priced at the open
- Annualised volatility
  - close-to-close 16.6%
  - open-to-close 13.6%
  - overnight 10.1%
- For daily risk we use **total** variance: $\mathrm{RV}_t + r^2_{\textbf{overnight},t}$
  - annualised mean: 16.4%
  - Alternatives: scale intraday RV up, or weight the two parts optimally Hansen & Lunde (2005)

MFM_ch9_overnight

▸ Appendix: day and night variance

How Precise Is Realised Variance?

- Feasible central limit theorem (CLT) for RV Barndorff-Nielsen & Shephard (2002)
  - $\sqrt{M}\,(\mathrm{RV}_t - \mathrm{IV}_t)/\sqrt{2\,\mathrm{RQ}_t} \to N(0,1)$
  - $\mathrm{IQ}_t = \int_{t-1}^{t} \sigma^4(s)\,ds$: the integrated quarticity, which sets the variance of RV
  - IQ_t is estimated by the realised quarticity $\mathrm{RQ}_t = \frac{M}{3}\sum_i r_{t,i}^4$
- Standard error: $\mathrm{se}(\mathrm{RV}_t) = \sqrt{\frac{2}{3}\sum_i r_{t,i}^4}$
- 95% confidence interval (CI): $\mathrm{RV}_t \exp\left(\pm 1.96\,\mathrm{se}(\mathrm{RV}_t)/\mathrm{RV}_t\right)$
  - on the log scale, so always positive
  - With constant volatility, $\mathrm{se}/\mathrm{RV} \approx \sqrt{2/M}$: 16% for $M = 78$
- The limiting variance $2\,\mathrm{IQ}_t$ is itself random (a mixed Normal limit)
  - replacing it by $2\,\mathrm{RQ}_t$ is legitimate because the convergence is **stable**
  - formal statement in the Appendix

Why Not Use Every Tick? (1/2)

- **Question**: why not use every trade?
  - more returns shrink the relative error of RV ($\sqrt{2/M}$)
- Observed prices are not the efficient price: $p_i^{\mathbf{obs}} = p_i^* + \varepsilon_i$
  - so $r_i = r_i^* + \varepsilon_i - \varepsilon_{i-\mathbf{1}}$
  - p_i^*: the efficient (true) log price, ε_i: the noise in the observed price, r_i^*: the efficient return
  - Sources of ε: bid–ask bounce, tick size (price discreteness), stale quotes (Chapter 10)

Why Not Use Every Tick? (2/2)

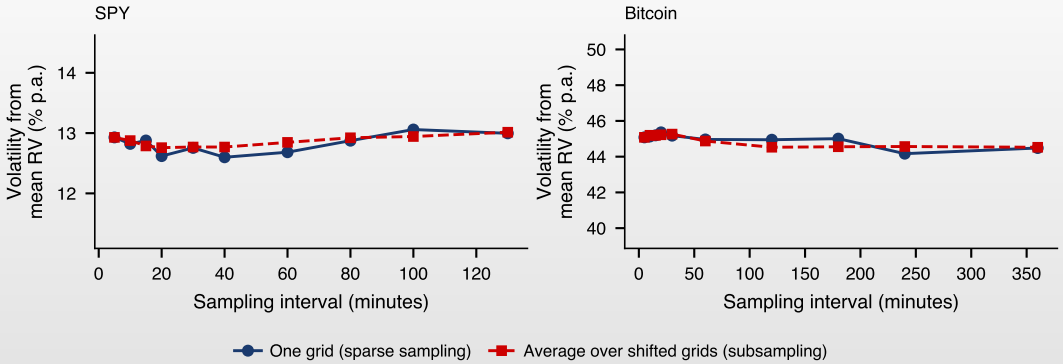
- With i.i.d. noise of variance ω^2 and n returns per day Zhang, Mykland & Aït-Sahalia (2005), Hansen & Lunde (2006)
  - $\mathbb{E}[\mathrm{RV}_t^{(n)} \mid \sigma] = \mathrm{IV}_t + 2n\omega^2$ (zero drift, no leverage): each squared return carries $2\omega^2$ of noise, so the bias **grows** with the sampling frequency
  - Adjacent returns share ε_{i-1} with opposite signs: $\mathrm{cov}(r_i, r_{i-1}) = -\omega^2$
- Trade-off: $\mathbb{E}[(\mathrm{RV}^{(n)} - \mathrm{IV})^2] \approx 2\,\mathrm{IQ}/n + 4n^2\omega^4$
  - discretisation error plus squared bias
  - MSE (mean squared error) optimal number of returns $n^* = (\mathrm{IQ}/(4\omega^4))^{1/3}$ Bandi & Russell (2008): more returns cut the first term, but raise the second; derivation in the Appendix
- Numbers: bias 0.016 at 5 minutes ($n = 78$), 4.680 at 1 second
  - $\mathrm{IV} = 1\%^2$ with constant volatility within the day, so $\mathrm{IQ} = \mathrm{IV}^2 = 1\ (\%^4)$
  - $\omega = 0.01\%$, so $\omega^2 = 10^{-4}\%^2$
- Optimal number of returns: $n^* = 292$ (one return every 80 s)

Appendix: simulated signature plot

Why 5 Minutes?

- **Question**: which frequency, when n^* depends on ω^2 and IQ_t?
  - both are unknown and different across assets
- The optimal frequency falls fast as the noise grows
  - $n^* = 292$ with $\omega = 0.01\%$
  - $n^* = 14$ (one return about every 29 minutes) with $\omega = 0.1\%$
- **Sparse sampling** at 5 minutes ($n = 78$ in a 6.5-hour session): a robust compromise Liu, Patton & Sheppard (2015)
  - the bias $2n\omega^2$ stays small, and the relative discretisation error is $\sqrt{2/78} = 16\%$
  - the cost: most trades are discarded
- The course data are 5-minute bars, the finest grid available
  - the signature plot on real data checks whether 5 minutes is sparse enough

The Signature Plot on Real Data (1/2)



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The Signature Plot on Real Data (2/2)

- Annualised volatility from mean RV by sampling interval
  - one grid vs the average over all shifted grids
- Flat in the sampling interval
  - SPY 12.9% at 5 minutes, 12.8% at 30, 13.0% at 130
  - Bitcoin 45.1% at 5 minutes, 44.5% at 6 hours
  - First-order autocorrelation of 5-minute returns: -0.008 (SPY), 0.005 (Bitcoin): no visible noise at 5 minutes
- Limit of this dataset: 5-minute bars can diagnose 5-minute and coarser sampling only
  - they say nothing about noise below 5 minutes
- On 5-minute SPY data, ratios to RV
  - two-scale RV / RV $= 0.97$, correlation with RV 0.89
  - realised kernel / RV $= 0.95$, correlation with RV 0.85
- Consistent with Liu, Patton & Sheppard (2015): almost 400 estimators on 31 assets rarely beat 5-minute RV
  - noise-robust estimators pay off when noise is large relative to the signal (tick or second-level data, illiquid assets)

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Prices Jump: Bipower Variation (1/2)



Ole E. Barndorff-Nielsen, 2007

Photo: Thomas Steiner (2007); CC BY-SA 2.5; Wikimedia Commons

- **Question**: how much of a day's variance comes from jumps, and how do we separate jumps from diffusion?
- Add jumps to the model: $dp = \mu\, ds + \sigma\, dW + dJ$
  - J: a jump process (news, Chapter 11)
  - Quadratic variation now has two parts: $\mathrm{RV}_t \to \mathrm{IV}_t + \sum_j \kappa_j^2$, with κ_j the jump sizes in day t
  - Why separate them? Jumps are less persistent than diffusive volatility, and they price differently (options, Chapter 12)

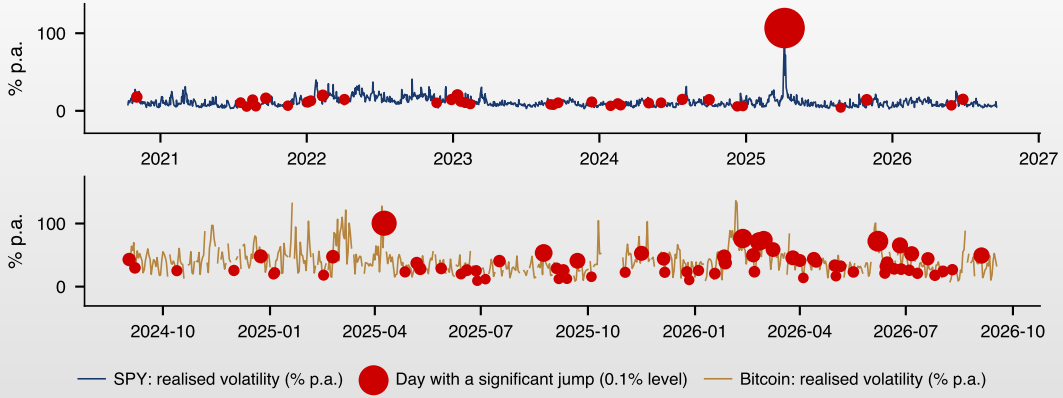
Prices Jump: Bipower Variation (2/2)

- **Bipower variation** Barndorff-Nielsen & Shephard (2004): products of adjacent absolute returns
  - $\mathrm{BV}_t = \mu_1^{-2} \frac{M}{M-1} \sum_{i=2}^{M} |r_{t,i}|\,|r_{t,i-1}| \to \mathrm{IV}_t$
  - $\mu_1 = \mathbb{E}|Z| = \sqrt{2/\pi} = 0.798$ for $Z \sim N(0,1)$, so $\mu_1^{-2} = \pi/2 = 1.571$; $\frac{M}{M-1}$: finite-sample normalisation
- A jump enters one return, which is multiplied by a small neighbour
  - its effect on BV vanishes as M grows
- The jump part of the day is $\max(\mathrm{RV}_t - \mathrm{BV}_t, 0)$
  - Example, $M = 4$ returns $0.1, 0.1, 0.1, 1.0$ (%): $\mathrm{RV} = 1.03$, $\mathrm{BV} = \frac{\pi}{2} \cdot \frac{4}{3}(0.01 + 0.01 + 0.10) = 0.25$, jump part 0.78 (%^{2})

A Test for Jumps

- Ratio statistic Huang & Tauchen (2005), Barndorff-Nielsen & Shephard (2006)
  - $z_t = \dfrac{(\mathrm{RV}_t - \mathrm{BV}_t)/\mathrm{RV}_t}{\sqrt{\theta\, M^{-1} \max\left(1, \mathrm{TQ}_t/\mathrm{BV}_t^2\right)}}$
  - $\theta = \pi^2/4 + \pi - 5 = 0.609$; tripower quarticity $\mathrm{TQ}_t = M\mu_{4/3}^{-3} \frac{M}{M-2} \sum_i |r_{t,i}|^{4/3} |r_{t,i-1}|^{4/3} |r_{t,i-2}|^{4/3}$, $\mu_{4/3} = \mathbb{E}|Z|^{4/3}$
  - Numerator: the relative jump share $(\mathrm{RV}_t - \mathrm{BV}_t)/\mathrm{RV}_t$; denominator: its standard error
  - θ: the asymptotic variance constant; TQ estimates IQ_t robustly to jumps
- Under no jumps $z_t \to N(0,1)$
  - a jump makes $\mathrm{RV} > \mathrm{BV}$ and z_t large and positive
- We use $\alpha = 0.1\%$ (critical value 3.09) for the jump days plotted on the next slide
- Significant day: jump part $J_t = \mathrm{RV}_t - \mathrm{BV}_t$
  - continuous part $C_t = \mathrm{RV}_t - J_t$
  - otherwise $J_t = 0$

Jump Days: SPY and Bitcoin (1/2)



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Jump Days: SPY and Bitcoin (2/2)

- Red points: days with a significant jump at the 0.1% level
  - point size proportional to the jump part J_t
- SPY: 35 jump days of 1 489 (2.4%)
  - jumps are 2.8% of total realised variance
  - on a jump day the jump is on average 35% of that day's RV
  - largest statistic: 2021-08-27 ($z = 5.2$), 10:00–10:05: the Jackson Hole speech of the Fed chair
- Bitcoin: 67 jump days of 659 (10.2%), but only 1.8% of variance
  - many small jumps on quiet days (weekends), where one isolated move dominates the day
- The turbulent days of April 2025 are mostly **not** jump days
  - many large returns in a row raise BV as well

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Many Days, Many Tests: Spurious Jumps

- 1 489 daily tests at 5%: 235 rejections; which of them are false?
  - against 74 expected under no jumps
- Part of the excess is intraday seasonality (Further Topics)
  - dividing returns by the periodicity factor f_i Boudt, Croux & Laurent (2011) already cuts the 5% rejections from 235 to 167
- Benjamini & Hochberg (1995): sort the p-values and reject the k smallest
  - k: the largest index with $p_{(k)} \leq kq/N$; $p_{(k)}$: k-th smallest p-value, N: number of tests, q: target FDR
  - here each day's p-value is the one-sided $1 - \Phi(z_t)$ of the ratio statistic
  - the FDR, $\mathbb{E}$[false/rejections], stays $\leq q$ (false/rejections $:= 0$ when nothing is rejected)
  - The guarantee needs valid p-values that are independent or positively dependent Benjamini & Yekutieli (2001)
  - daily jump tests are only approximately so

▸ Appendix: BH example, five tests

Jump Counts under Multiplicity Control

- SPY, 1 489 days
  - FDR at 5%: 42 jump days; at 1%: 9
  - Bonferroni (family-wise error rate, FWER, 5%): 7
  - FDR 5% after the periodicity correction: 17
  - from 235 unadjusted rejections to 17: most rejections do not survive the two corrections for seasonality and multiplicity
- Bajgrowicz, Scaillet & Treccani (2016): without multiplicity control most detected jumps are spurious
  - the true ones are rare and cluster on news
- Christensen, Oomen & Podolskij (2014): most 5-minute "jumps" are steep but continuous bursts of volatility
  - at tick frequency (fact or friction)
- A daily test answers "did day t contain a jump?"
  - locating the jump within the day needs an intraday test (Lee–Mykland, Appendix)

Check Yourself: Measurement, Noise and Jumps (1/2)

- Why does the standard error of RV use fourth powers of returns?
  - The variance of RV depends on integrated quarticity IQ_t, estimated by $\mathrm{RQ}_t = \frac{M}{3} \sum_i r_{t,i}^4$
- Why is SPY's daily risk understated if you use intraday RV alone?
  - Overnight returns bring 38% of SPY daily variance
- Why does sampling every tick overstate variance?
  - Every squared return adds $2\omega^2$ of noise, so the bias $2n\omega^2$ grows with the number of returns

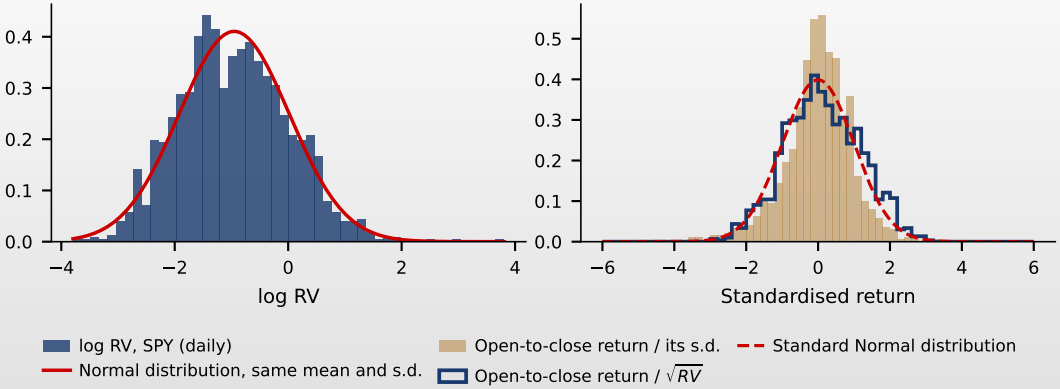
Check Yourself: Measurement, Noise and Jumps (2/2)

- What does a flat signature plot tell you?
  - It shows that noise is negligible at 5 minutes and coarser
- What does it not tell you?
  - It says nothing about noise below 5 minutes
- Why does bipower variation stay close to IV on a jump day while RV does not?
  - The jump enters a single return: squared in RV, multiplied by a small neighbour in BV
- With 1 489 daily jump tests, why is "$p < 0.05$" not evidence of a jump?
  - At 5%, 1 489 jump-free days would give about 74 false rejections
- What does the FDR control?
  - The FDR controls the expected share of false rejections among rejections

What Regularities Does Realised Volatility Show?

- **Question**: which properties of RV must a forecasting model reproduce?
- Distribution: is RV roughly lognormal? If so, we model $\ln \mathrm{RV}$
- Returns divided by $\sqrt{\mathrm{RV}_t}$: how much of the fat tails is due to time-varying volatility?
- Persistence: how slowly does the autocorrelation of $\ln \mathrm{RV}$ decay?
- Implied volatility: does the VIX forecast realised volatility over the next month?

Distributions: log RV and Standardised Returns (1/2)



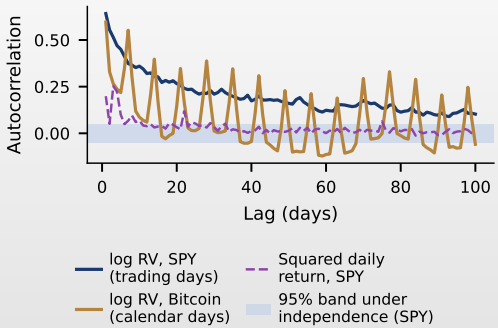
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Distributions: log RV and Standardised Returns (2/2)

- Left: daily log RV of SPY and the Normal density
- Right: open-to-close returns scaled
  - by their standard deviation (filled)
  - by $\sqrt{\mathrm{RV}_t}$ (outline)
- SPY skewness and kurtosis (the Normal distribution: 0 and 3)
  - RV: skewness 20.0, kurtosis 550
  - log RV: skewness 0.29, kurtosis 3.20
  - Bitcoin log RV: skewness -0.27, kurtosis 3.33: model volatility in logs
- Open-to-close returns: kurtosis 15.73, and 2.65 after dividing by $\sqrt{\mathrm{RV}_t}$
  - standard deviation after dividing: 1.05
  - Bitcoin: from 7.57 to 2.77
- Heavy tails of daily returns (Chapter 1) come mostly, ex post, from changing volatility
  - approximately a **mixture** of Normal distributions with variances IV_t
  - exact only without leverage and jumps (section Realised Variance)

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Persistence of Realised Volatility (1/2)



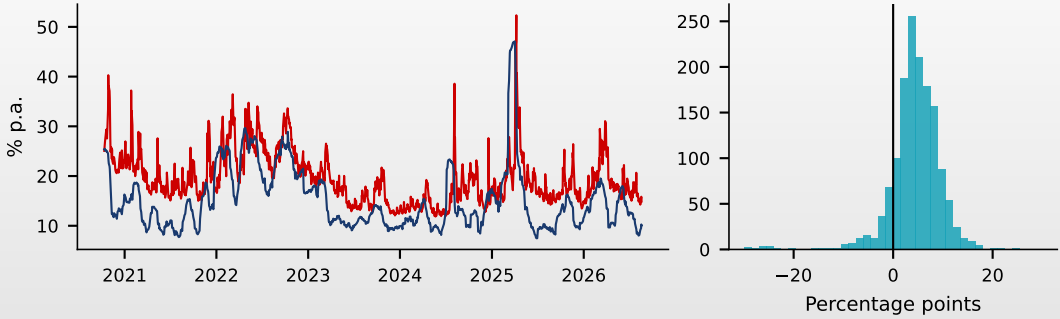
- Autocorrelations up to 100 days
  - log RV of SPY (total variance) and Bitcoin
  - squared daily SPY returns
- SPY log RV: **highly persistent** dependence
  - 0.64 at lag 1, 0.24 at one month, still 0.10 at 100 days
  - an autoregressive AR(1) model would decay geometrically: $0.64^{22} \approx 0.00005$
  - a sample autocorrelation function (ACF) is consistent with long memory but cannot prove it

Persistence of Realised Volatility (2/2)

- Squared daily returns show the same persistence, hidden by noise
  - autocorrelation at lag 1: 0.20
- Bitcoin: peaks at 7, 14, 21, ... days: a **weekly cycle**
  - weekend RV is only 0.41 of weekday RV (Saturday 27% vs Wednesday 48% p.a.)
  - no equity or macro news, fewer institutional traders

MFM_ch9_acf

Implied vs Realised Volatility (1/2)



MFM_ch9_vix

Implied vs Realised Volatility (2/2)

- Left: VIX and SPY realised volatility over the next 21 trading days
- Right: the gap between them
- VIX mean 19.52% vs 15.15% realised afterwards
- VIX was higher on 88% of 1 468 days, by 4.37 percentage points on average
- **Variance risk premium** (VRP) Carr & Wu (2009), Bollerslev, Tauchen & Zhou (2009)
  - $\mathrm{VRP}_t = \mathbb{E}_t^{\mathbb{Q}}[\mathrm{RV}_{t,t+21}] - \mathbb{E}_t^{\mathbb{P}}[\mathrm{RV}_{t,t+21}]$: risk-neutral minus physical expectation of realised variance over the next 21 trading days; VIX^2 approximates the first term (Chapter 12)
  - the chart shows an **ex-post** spread, a noisy proxy of this ex-ante premium
  - negative spread (March–April 2025): a forecast error, not a negative premium

MFM_ch9_vix

Appendix: VIX as a forecast

The HAR-RV Model (1/2)

- **Question**: how do we forecast tomorrow's RV when the autocorrelation of $\ln \mathrm{RV}$ decays very slowly?
  - the autocorrelation is still 0.10 at 100 days
- Corsi (2009): $\mathrm{RV}_{t+1} = \beta_0 + \beta_d \mathrm{RV}_t + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)} + \varepsilon_{t+1}$
  - $\mathrm{RV}_t^{(w)} = \frac{1}{5} \sum_{j=0}^{4} \mathrm{RV}_{t-j}$ (week), $\mathrm{RV}_t^{(m)} = \frac{1}{22} \sum_{j=0}^{21} \mathrm{RV}_{t-j}$ (month); Bitcoin trades every day, so we use 7 and 30 days
  - β_d, β_w, β_m: weights of yesterday's RV and of last week's and last month's average RV; β_0: constant; ε_{t+1}: forecast error
- Heterogeneous market hypothesis Müller et al. (1997): daily, weekly and monthly traders
  - the "H" of HAR; HAR-RV: HAR model of realised variance
  - a few components with different persistence mimic long memory

The HAR-RV Model (2/2)

- An AR(22) with only 3 free slopes
  - weight on lag 1: $\beta_d + \beta_w/5 + \beta_m/22$
  - weight on lags 2–5: $\beta_w/5 + \beta_m/22$
  - weight on lags 6–22: $\beta_m/22$
- Estimation by ordinary least squares (OLS)
  - errors are heteroskedastic and autocorrelated, so we use Newey–West standard errors
  - HAC: heteroskedasticity- and autocorrelation-consistent

The log-HAR Model

- Why logs
  - $\ln \mathrm{RV}$ is close to Normal, whereas RV is very heavy-tailed (slide *Distributions*)
  - the forecast $\exp(\cdot)$ is always positive
  - extreme days no longer dominate the OLS fit
- $\ln \mathrm{RV}_{t+1} = \beta_0 + \beta_d \ln \mathrm{RV}_t + \beta_w \overline{\ln \mathrm{RV}}_t^{(w)} + \beta_m \overline{\ln \mathrm{RV}}_t^{(m)} + \varepsilon_{t+1}$
  - with $\overline{\ln \mathrm{RV}}_t^{(w)} = \frac{1}{5} \sum_{j=0}^{4} \ln \mathrm{RV}_{t-j}$ and $\overline{\ln \mathrm{RV}}_t^{(m)} = \frac{1}{22} \sum_{j=0}^{21} \ln \mathrm{RV}_{t-j}$
  - geometric means of variances
- Variance forecast: $\exp(\hat{y} + \hat{s}^2/2)$
  - $\hat{y}$: the fitted $\ln \mathrm{RV}$; $\hat{s}^2$: the residual variance
  - the conditional mean if the errors are $N(0, s^2)$ (lognormal approximation)
  - for non-Normal errors use the smearing factor $\frac{1}{T} \sum_t e^{\hat{\varepsilon}_t}$ Duan (1983)
  - with heteroskedastic errors both are approximations

HAR Estimates: SPY and Bitcoin

	β_0	β_d	β_w	β_m	R^2
SPY, HAR (levels)	0.273	0.399	0.222	0.114	0.31
	(0.064)	(0.089)	(0.096)	(0.053)	
SPY, log-HAR	−0.079	0.362	0.337	0.167	0.47
	(0.027)	(0.037)	(0.048)	(0.044)	
Bitcoin, HAR (levels)	1.496	0.412	0.409	−0.065	0.34
	(0.412)	(0.089)	(0.122)	(0.097)	
Bitcoin, log-HAR	0.223	0.380	0.465	−0.022	0.36
	(0.094)	(0.041)	(0.080)	(0.098)	

- Full samples
  - SPY: total daily variance (intraday RV plus squared overnight return), 1 467 days
  - Bitcoin: 24-hour RV, 547 days on which all three HAR regressors are available
- Newey–West standard errors in brackets

Reading the HAR Estimates

- SPY: all three components are significant
  - Persistence $\hat{\beta}_d + \hat{\beta}_w + \hat{\beta}_m = 0.87$: shocks fade slowly, as in GARCH (Chapter 5)
- $R^2 = 0.47$ for log variance vs 0.31 for variance in levels
  - not comparable: the dependent variables differ
  - the out-of-sample losses on a common target (section Forecasting: HAR vs GARCH) decide
- Bitcoin: daily and weekly components matter
  - the monthly coefficient is close to 0 (-0.022, SE 0.098)
  - Short sample (two years) and a strong weekly cycle: the week absorbs most of the memory

▸ Appendix: lag weights, extensions

HARQ and SHAR

- **HARQ** Bollerslev, Patton & Quaedvlieg (2016): yesterday's RV is measured with error, and the error attenuates the daily slope
  - $\mathrm{Var}(\mathrm{RV}_t - \mathrm{IV}_t) \approx 2\,\mathrm{IQ}_t / M$ (Appendix)
  - $\mathrm{RV}_{t+1} = \beta_0 + (\beta_d + \beta_Q \sqrt{\mathrm{RQ}_t})\,\mathrm{RV}_t + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)} + \varepsilon_{t+1}$
  - $\beta_Q < 0$: yesterday's RV gets less weight on days with high realised quarticity, that is, with large measurement error
- **SHAR** Patton & Sheppard (2015): realised semivariances split RV by the sign of the return (Appendix)
  - $\mathrm{RS}_t^+ = \sum_i r_{t,i}^2 1\{r_{t,i} > 0\}$, $\mathrm{RS}_t^- = \sum_i r_{t,i}^2 1\{r_{t,i} < 0\}$, so $\mathrm{RS}_t^+ + \mathrm{RS}_t^- = \mathrm{RV}_t$
  - $\mathrm{RV}_{t+1} = \beta_0 + \beta^+ \mathrm{RS}_t^+ + \beta^- \mathrm{RS}_t^- + \beta_w \mathrm{RV}_t^{(w)} + \beta_m \mathrm{RV}_t^{(m)} + \varepsilon_{t+1}$
  - why: "bad" volatility, from negative returns, predicts more future volatility than "good" volatility, so $\beta^- > \beta^+$

Which Model Forecasts Tomorrow's Variance Best?

- Target: tomorrow's daily variance, forecast only with data available the evening before
  - SPY: total variance, 3 January 2022 to 18 September 2026 (1 180 days)
  - Bitcoin: 1 March 2025 onwards (422 days)
- Five models
  - log-HAR and HAR on realised variance from intraday data, re-estimated daily on an expanding window
  - GARCH(1,1) with Student-t errors on daily returns since 2000 (SPY) or 2014 (Bitcoin), re-estimated every 21 days (GARCH: Bollerslev (1986); Student-t errors: Bollerslev (1987))
  - EWMA (exponentially weighted moving average) on daily returns with decay factor $\lambda = 0.94$ (Chapter 5)
  - "yesterday's RV" (random walk)

Evaluating Against a Noisy Target

- True variance is unobserved
  - we compare forecasts F_t with the proxy RV_t
  - A loss is **robust** if the ranking by *expected* loss with a proxy that is conditionally unbiased (given the forecasts' information) equals the ranking with true variance Patton (2011)
- MSE $= (\mathrm{RV}_t - F_t)^2$: robust
  - measures errors in variance units
  - is dominated by the most volatile days
- QLIKE $= \dfrac{\mathrm{RV}_t}{F_t} - \ln \dfrac{\mathrm{RV}_t}{F_t} - 1 \geq 0$: robust
  - depends only on the ratio
  - penalises under-prediction more
  - Where QLIKE comes from: for $r \sim N(0, F)$, $-2 \ln$ density is $\ln F + r^2/F$ plus a constant; replacing r^2 by RV_t and subtracting the minimum, reached at $F_t = \mathrm{RV}_t$, gives the formula above
- Common mistake: MAE, or MSE on volatility instead of variance, is **not** robust
  - MAE: mean absolute error
  - Andersen & Bollerslev (1998): against r_t^2, GARCH forecasts explained a few % of the variation; against RV the same forecasts explain far more: the proxy was noisy, not the model

Comparing Two Forecasts: Diebold–Mariano

- Diebold & Mariano (1995) (DM) test: $t = \bar{d}/\mathrm{se}_{\mathrm{NW}}(\bar{d}) \to N(0,1)$
  - loss difference: $d_t = L(\mathrm{RV}_t, F_t^A) - L(\mathrm{RV}_t, F_t^B)$
  - $L(\mathrm{RV}_t, F_t)$: the loss (QLIKE or MSE) of forecast F_t; $\bar{d}$: mean of d_t; $\mathrm{se}_{\mathrm{NW}}$: Newey–West standard error
  - $t < 0$: model A has the lower loss
  - HAC standard error because d_t is autocorrelated (Chapter 8)
- What drives t: the mean gap **and** how stable the gap is from day to day
  - Check, ignoring autocorrelation, $T = 100$ days: a gap with mean 0.01 and standard deviation 0.02 gives $t = 0.01/(0.02/\sqrt{100}) = 5$
  - a gap with mean 0.03 and standard deviation 0.30 gives $t = 1$
  - two forecasts that move together have a stable gap: a small mean difference can then be highly significant

Calibration: Mincer–Zarnowitz

- Mincer & Zarnowitz (1969) (MZ): $\mathrm{RV}_t = a + b\,F_t + u_t$
  - regress the realised target on the forecast
  - a: intercept, b: slope, u_t: error
  - an unbiased forecast has $a = 0$, $b = 1$
  - $b < 1$: forecasts too extreme
  - $b > 1$: forecasts too timid: realised variance moves more across days than the forecast does
  - Check: a forecast of 2 on calm days and 4 on busy days, with realised means 1.5 and 4.5, has $b = (4.5 - 1.5)/(4 - 2) = 1.5 > 1$ and $a = 1.5 - 1.5 \times 2 = -1.5$: it under-separates the two kinds of day
  - Joint Wald test with Newey–West covariance
  - R^2 measures how much of the variation the forecast explains
  - R^2 is capped by the noise in RV: even the true variance would not reach 1 Andersen & Bollerslev (1998)

Results: SPY, 2022–2026 (1/2)

Model	QLIKE	$\mathrm{DM}_{\mathrm{QLIKE}}$ t (p)	MSE ($\%^4$)	$\mathrm{DM}_{\mathrm{MSE}}$ p	MZ b	MZ R^2
log-HAR	0.310	–	4.15	–	1.16	0.30
HAR	0.326	−1.95 (0.052)	4.48	0.117	0.75	0.26
GARCH(1,1)-t	0.343	−2.30 (0.021)	4.87	0.188	0.70	0.20
EWMA	0.393	−4.02 (< 0.001)	5.14	0.050	0.74	0.14
Yesterday's RV	0.502	−5.95 (< 0.001)	5.41	0.060	0.54	0.29

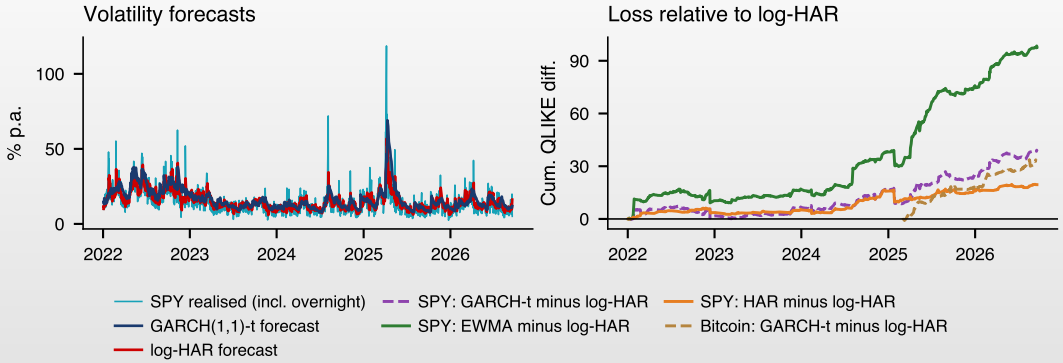
- 1 180 one-day-ahead forecasts of total daily variance
  - DM against log-HAR: $d_t = L(\text{log-HAR}) - L(\text{model})$ with L = QLIKE or MSE, Newey–West
  - $t < 0$ favours log-HAR
- log-HAR has the lowest QLIKE and MSE
  - QLIKE is 10% below GARCH(1,1)-t
  - Against EWMA and yesterday's RV the gain is highly significant; against HAR in levels only marginally
- MSE differences are not significant at 5% (EWMA borderline, $p = 0.050$)
  - a few crisis days (April 2025) dominate the squared errors

Results: SPY, 2022–2026 (2/2)

- Mincer–Zarnowitz: $a = 0, b = 1$ is not rejected for log-HAR ($p = 0.554$)
  - its R^2 is higher than GARCH's
  - log-HAR $b > 1$: if anything slightly too timid (0.66 standard errors from 1, SE 0.25)
  - GARCH $b < 1$: slightly too extreme, also not significant
- Yesterday's RV has a high R^2 but $b = 0.54$
  - right direction, far too extreme

▸ Appendix: Bitcoin results

Forecasts over Time (1/2)



MFM_ch9_forecasts

Forecasts over Time (2/2)

- Left: SPY realised volatility (with overnight) and the two forecasts, annualised
- Right: cumulative QLIKE loss of each model minus log-HAR (rising = log-HAR better)
- After a shock both forecasts jump, but GARCH stays high longer
  - GARCH persistence is close to 1, while log-HAR returns faster
  - April 2025: GARCH above 40% p.a. until late April
  - realised volatility below 20% from mid-April
- The cumulative loss difference rises steadily, with jumps at turbulent episodes
  - the gain is not driven by one day
- Of the three models plotted against log-HAR, EWMA loses most
  - a fixed $\lambda = 0.94$ reacts too much to single days

MFM_ch9_forecasts

▸ Appendix: Mincer–Zarnowitz chart

Which Models Survive? The Model Confidence Set

Model	QLIKE	MCS p-value	Eliminated at step	In MCS (90%)
log-HAR	0.310	1.000	–	✓
SHAR	0.320	0.224	–	✓
HAR	0.326	0.014	3	
HARQ	0.344	0.095	4–5	
GARCH(1,1)-t	0.343	0.095	4–5	
EWMA	0.393	0.012	2	
Yesterday's RV	0.502	< 0.001	1	

- Model confidence set (MCS) Hansen, Lunde & Nason (2011)
  - on the five models plus SHAR and HARQ
  - test equal predictive ability, drop the worst model, repeat
  - range statistic $T_R = \max_{i,j} |\bar{d}_{ij}| / \widehat{\mathrm{se}}(\bar{d}_{ij})$, with $\bar{d}_{ij}$ the mean loss gap between surviving models i and j
  - distribution of T_R: block bootstrap with 20-day blocks and 10 000 replications, so the smallest reportable p-value is of order 10^{-4}
  - the surviving set contains the best model with asymptotic probability $\geq 90\%$
- SPY, 1 180 days from 2022: the 90% set is {log-HAR, SHAR}
- A multiple-comparison procedure
  - many pairwise DM tests without size control would overstate the evidence (Chapter 8)

▸ Appendix: reading MCS p-values

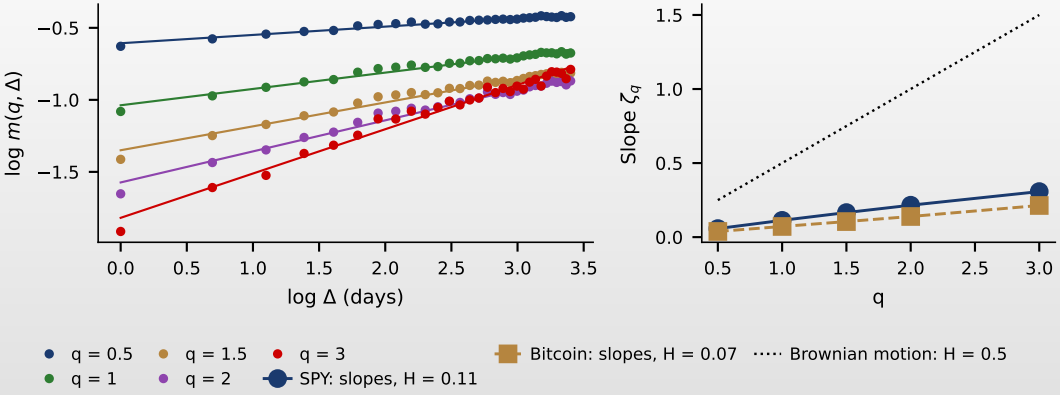
Is Volatility Rough?

- **Question**: are log-volatility paths rougher than those of a Brownian motion?
- Gatheral, Jaisson & Rosenbaum (2018): log volatility behaves like a **fractional Brownian motion**
  - Hurst exponent $H \approx 0.1$
  - $H = 0.5$: ordinary Brownian motion
  - $H < 0.5$: increments are negatively correlated, paths are rougher
  - $H > 0.5$: smoother, trending
- Rough models reproduce the shape of the implied volatility smile
  - for short maturities (Chapter 12)

Fractional Brownian Motion and the Scaling Law

- Fractional Brownian motion B_H, $0 < H < 1$: a Gaussian process with $B_H(0) = 0$ and mean 0
  - $\mathrm{Cov}(B_H(t), B_H(s)) = \frac{1}{2}\left(|t|^{2H} + |s|^{2H} - |t-s|^{2H}\right)$
  - $H = 1/2$: the covariance is $\min(t, s)$, ordinary Brownian motion
  - increments: $B_H(t+\Delta) - B_H(t) \sim N(0, \Delta^{2H})$ for every t
- Model $\ln \sigma_t = \text{const} + \nu B_H(t)$
  - the change over Δ days is $\nu \Delta^H Z$ in distribution, $Z \sim N(0,1)$
  - Scaling law: $m(q, \Delta) = \frac{1}{T}\sum_t |\ln \sigma_{t+\Delta} - \ln \sigma_t|^q$, with $\sigma_t = \sqrt{\mathrm{RV}_t}$, estimates $\nu^q \, \mathbb{E}|Z|^q \, \Delta^{qH}$, so the slope of $\ln m(q, \Delta)$ on $\ln \Delta$ is $\zeta_q = qH$
- Estimate: slopes ζ_q for $q \in \{0.5, 1, 1.5, 2, 3\}$
  - then regress ζ_q on q through the origin: $\hat{H} = \sum_q q\,\zeta_q / \sum_q q^2$
- Check with $q = 2$
  - for Brownian motion $m(2, \Delta) = \nu^2 \Delta$, so doubling Δ doubles m and $\zeta_2 = 1 = 2 \times 0.5$
  - with $H = 0.1$, doubling Δ multiplies $m(2, \Delta)$ only by $2^{0.2} \approx 1.15$

Scaling of Log Volatility Increments (1/2)



MFM_ch9_rough

Scaling of Log Volatility Increments (2/2)

- Left: $\ln m(q, \Delta)$ against $\ln \Delta$ for SPY, $\Delta = 1, \ldots, 30$ days
- Right: slopes ζ_q against q
- The log–log points lie close to straight lines and the slopes are close to linear in q
  - consistent with the scaling law, which the plot alone cannot establish
  - $\hat{H} = 0.11$ for SPY (block bootstrap 95% CI $[0.08; 0.13]$), 0.07 for Bitcoin: far below 0.5
  - Bitcoin uses only observed pairs at each calendar lag; interpolating the missing days gives 0.08
- The catch: RV measures volatility with error
  - the next slides show how measurement error lowers $\hat{H}$

MFM_ch9_rough

Rough or Noisy? The Identification Problem

- The estimate $H \approx 0.1$ comes from RV, a noisy proxy of the latent σ_t
  - Cont & Das (2024): the scaling estimator applied to realised volatility finds roughness even when spot volatility is smooth: an estimation artefact
  - Bolko, Christensen, Pakkanen & Veliyev (2023) (generalised method of moments, GMM, that models the RV error) and Fukasawa, Takabatake & Westphal (2022) (consistent under high-frequency asymptotics) still find $H < 0.5$
- Formal question: $\ln \mathrm{RV}_t^{1/2} = \ln \sigma_t + e_t$
  - $\ln \sigma_t$ is fractional of index H, e_t is the measurement error of RV
  - test H_0: $H = 1/2$ (classical diffusive volatility) against H_1: $H < 1/2$
  - Identification problem: a small H and a large $\mathrm{Var}(e_t)$ both flatten $m(2, \Delta)$ at short lags (next slide)
- Short-run roughness can coexist with long memory Bennedsen, Lunde & Pakkanen (2022)
- Stakes: the short-maturity skew and volatility forecasts depend on H (Chapter 12)

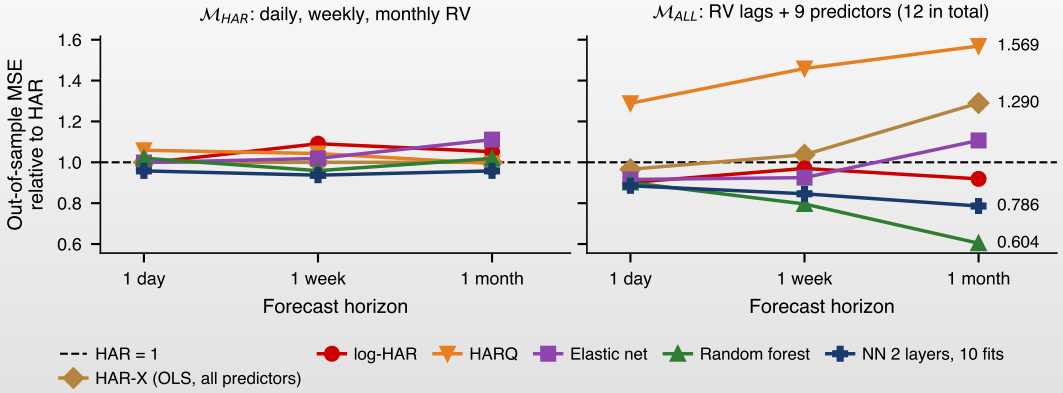
How Measurement Error Flattens the Scaling Law

- Observed increment = latent increment + error increment
  - $(\ln \sigma_{t+\Delta} - \ln \sigma_t) + (e_{t+\Delta} - e_t)$
  - if e_t is i.i.d. and independent of σ, the variances add: $\mathrm{Var}(e_{t+\Delta} - e_t) = 2\,\mathrm{Var}(e_t)$ for every $\Delta \geq 1$
  - hence $m(2, \Delta) = c\,\Delta^{2H} + k$, $c = \nu^2$, $k = 2\,\mathrm{Var}(e_t)$: a constant that does not grow with Δ
- Toy: $H = 0.5$, $c = 1$, $k = 1$
  - $m(2,1) = 1 + 1 = 2$, $m(2,2) = 2 + 1 = 3$: log–log slope $\ln(3/2)/\ln 2 = 0.58$, so $\hat{H} = 0.58/2 = 0.29$ instead of 0.5
  - at long lags the constant matters less: from $\Delta = 20$ to 40, slope $\ln(41/21)/\ln 2 = 0.97$, $\hat{H} = 0.48$
- Testable implication: under i.i.d. error, $\hat{H}$ rises when short lags are dropped
  - the mini-case in section AI for Scientific Discovery runs exactly this check on SPY

Case Study: Can Machine Learning Beat HAR?

- Christensen, Siggaard & Veliyev (2023): does off-the-shelf machine learning (ML) forecast daily realised variance better than the HAR family?
  - if so, why?
  - why ask: HAR Corsi (2009) is linear in three RV averages, while RV is right-skewed and persistent, so a nonlinear map, or predictors beyond past RV, may help
- Data: 29 of the 30 stocks of the Dow Jones Industrial Average (DJIA)
  - January 2001 – December 2017, $T = 4\,257$ days
  - 5-minute RV ($n = 78$)
- Information sets: $\mathcal{M}_{\text{HAR}}$ and $\mathcal{M}_{\text{ALL}}$
  - $\mathcal{M}_{\text{HAR}}$: daily, weekly, monthly RV
  - $\mathcal{M}_{\text{ALL}}$ adds 9 predictors: implied volatility, VIX, volume, momentum, macro and uncertainty indices

Case Study: Gains by Horizon (1/2)



MFM_ch9_ml_vs_har

Case Study: Gains by Horizon (2/2)

- Source: Christensen, Siggaard & Veliyev (2023)
  - out-of-sample MSE relative to HAR, averaged over 29 DJIA stocks (test set: the last 20% of 2001–2017)
  - left: RV lags only; right: all 12 predictors
  - HAR-X: OLS on all predictors
- RV lags only: the six models shown stay between 0.937 and 1.111 times the HAR MSE
  - at all three horizons
  - three RV averages leave little for a flexible model to learn
- All 12 predictors: the gain of trees and networks grows with the horizon
  - the authors attribute this to the higher persistence of the ML fits
  - random forest $0.901 \to 0.796 \to 0.604$
  - network with 2 hidden layers, averaged over 10 fits (NN_2^{10}): $0.885 \to 0.846 \to 0.786$
- Linear models with the same predictors deteriorate
  - HAR-X $0.966 \to 1.290$
  - HARQ $1.289 \to 1.569$
  - Among the linear models, only log-HAR stays below 1 at all horizons (0.901, 0.970, 0.919)

MFM_ch9_ml_vs_har

Appendix: ranking of 21 models

The Discovery Loop on the Roughness Question (1/2)

- Question: is $\hat{H} \approx 0.1$ a property of σ_t or of the RV error e_t?
- One pass of the loop with an AI assistant (reusable prompts in the Appendix)
  - Literature: tabulate the published estimates of H (estimator, frequency, treatment of the RV error)
  - check every DOI (digital object identifier) on Crossref
  - Competing hypotheses, each with its shape of $m(2, \Delta)$: fractional with $H = 0.1$; $H = 0.5$ plus i.i.d. error; two factors, one fast mean-reverting
  - Replicate the chapter's $\hat{H} = 0.11$ before extending
  - fix lag windows, measures and the fit with a constant noise term k in advance

The Discovery Loop on the Roughness Question (2/2)

- Checks the researcher owns
  - Proxy error: simulate $H = 0.5$ plus error of the same size; the procedure must not return $H \approx 0.1$
  - Overlapping increments invalidate OLS standard errors of ζ_q: use a block bootstrap
  - report every window tried
  - Assistants quote "$H \approx 0.1$ for all assets": check each number against the cited table

Mini-Case: Is $\hat{H}$ an Artefact of i.i.d. Error? (1/2)

- A typical assistant proposal: $\hat{H} \approx 0.1$ comes from i.i.d. measurement error in RV
  - testable implication: $\hat{H}$ **rises** when short lags are dropped

	$\Delta \in [1, 30]$	$[5, 30]$	$[10, 50]$	Fit with k, $[1, 50]$
Simulated $H = 0.5$ plus error	0.18	0.24	0.26	0.39
SPY, total RV	0.106	0.089	0.077	0.096
SPY, intraday RV	0.129	0.094	0.078	0.113

- First column replicates the chapter
- The simulated row shows the mechanism: $\hat{H}$ rises
  - the fit recovers most of the gap to 0.5
  - one path, not a power study

Mini-Case: Is $\hat{H}$ an Artefact of i.i.d. Error? (2/2)

- SPY: $\hat{H}$ **falls** with the window and the fitted k is 0
  - fit with k: $\hat{H} = 0.096$, block bootstrap 95% CI $[0.058; 0.268]$
- Interpretation: an illustrative diagnostic, inconsistent with the i.i.d.-error explanation
  - a formal rejection needs a simulation-calibrated test
  - the non-i.i.d. mechanism of Cont & Das (2024) is not tested by this design

MFM_ch9_ai_discovery_case

Your Turn: A Project Seed

- Question: does the roughness of SPY and Bitcoin volatility survive an estimator that models the RV measurement error?
- Replicate first (Seminar 9, B9)
  - $\hat{H} = 0.11$ (SPY) and 0.07 (Bitcoin) from the scaling law
  - $\hat{H} = 0.18$ for simulated $H = 0.5$ plus error
- Deliverables
  - A pre-registration (hypotheses, estimators, lag windows, tests, simulation design) dated before the first run
  - The GMM estimator of Bolko, Christensen, Pakkanen & Veliyev (2023) on the course RV series, with Monte Carlo size and power at $H \in \{0.1, 0.5\}$
  - Code, a two-page report with every specification tried, and a referee-style critique of your own result
- Log: github.com/danpele/MFM/tree/main/project
  - every prompt that shaped a design choice in `AI_USE.md`
  - every error the assistant made in `AI_ERRORS.md`

Which Tool When?

Situation	Suggested tool
Liquid asset, 1–5 minute bars	RV at 5 minutes, subsampled
Tick data, or an illiquid stock (Bucharest Stock Exchange, BVB)	Pre-averaging or flat-top realised kernel ($n^{-1/4}$; Parzen $n^{-1/5}$; Appendix: rates of convergence); check the signature plot
Daily risk of an exchange-traded asset	Total variance: RV plus squared overnight return
Separate news jumps from diffusion	BV or MedRV (median realised variance; Appendix: jump-robust estimators), ratio test with FDR control; Lee–Mykland for the jump time
Daily or weekly variance forecast	log-HAR (plus jumps, weekends, RQ); benchmark GARCH
Comparing forecasts	QLIKE and MSE on variance, Diebold–Mariano, Mincer–Zarnowitz, model confidence set

- Strongest result: out of sample, log-HAR has a lower QLIKE than GARCH(1,1)-t in both markets
  - MSE comparisons are inconclusive
- The 5-minute rule is justified only if the signature plot is flat at 5 minutes and coarser
- The jump counts depend on FDR control and on the periodicity correction

Check Yourself, and Next: Chapter 10

Retrieval questions

- Why is R^2 of log RV not comparable with R^2 of RV in levels?
  - slide *Reading the HAR Estimates*
- Why is QLIKE robust to a noisy RV proxy while MSE on volatility is not?
  - slide *Evaluating Against a Noisy Target*
- Why does the MCS state a set of models rather than a winner?
  - slide *Which Models Survive?*
- How can measurement error in RV make volatility look rough?
  - slide *How Measurement Error Flattens the Scaling Law*

Chapter 10: market microstructure

- Where does the noise ε of this chapter come from
  - limit order books, bid–ask spreads, tick sizes
- Liquidity, market impact and high-frequency trading, on equity and crypto exchanges

Two Markets, Two Clocks: SPY and Bitcoin

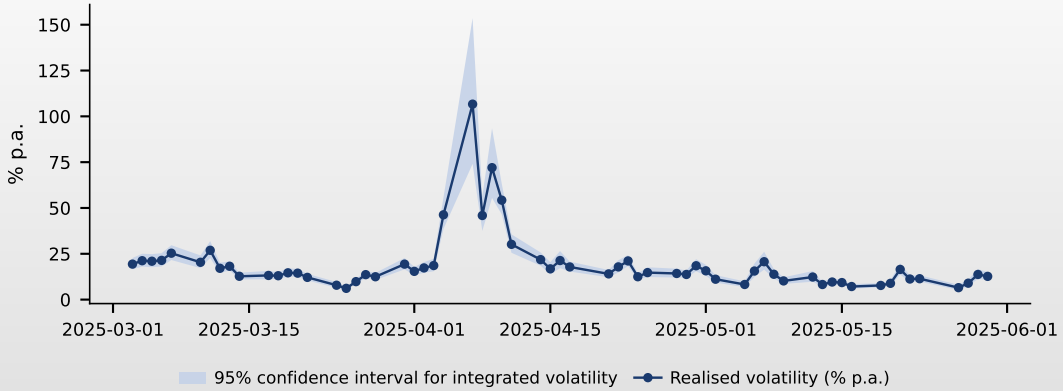
Series	Frequency	Period	Days
SPY (ETF)	5-minute bars, 09:30–16:00 New York time	2020–2026	1 489
Bitcoin	5-minute bars, 24-hour days (UTC)	2024–2026	659
SPY, Bitcoin	daily closes	2000/2014–2026	for GARCH
VIX	daily close	2020–2026	implied volatility

- SPY: 78 returns per full day, 43 on the 12 half-days
  - bars off the 5-minute grid removed
  - missing bars inside a day carry the last price
- Bitcoin: 288 returns per day
  - days with fewer than 95% of the bars dropped: 88 of 747 calendar days, mostly a data gap in October 2025

Adding Up Day and Night Variance

- SPY, annualised variances (%^{2} per year)
  - intraday RV: 12.9%, variance 167
  - total variance $\mathrm{RV}_t + r^2_{\text{overnight},t}$: 16.4%, variance 270
  - overnight part: $270 - 167 = 103$, i.e. $\sqrt{103} \approx 10.1\%$; share $103/270 = 38\%$
- Why not $13.6^2 + 10.1^2$?
  - the open-to-close 13.6% squares the whole session return, not each 5-minute return
  - $(\sum_i r_{t,i})^2 = \mathrm{RV}_t + 2\sum_{i<j} r_{t,i}\, r_{t,j}$: the cross products are zero only in expectation; in this sample they add about 17 ($13.6^2 \approx 184$)
  - close-to-close (16.6%) also contains the cross term $2\, r_{\text{overnight},t}\, r_{\text{open-close},t}$

Confidence Intervals for Daily Volatility (1/2)



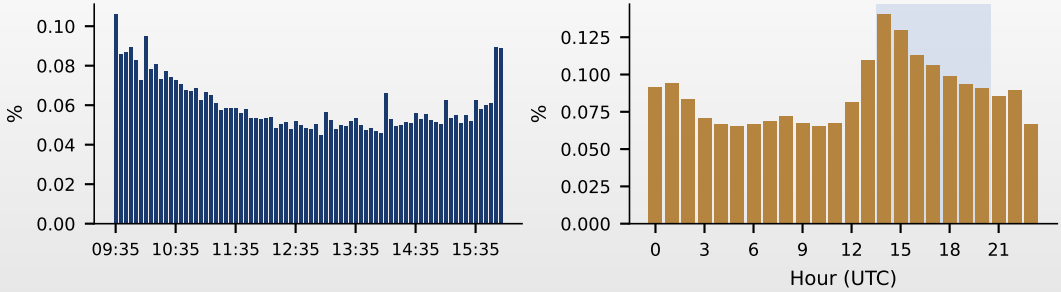
MFM_ch9_rv_ci

Confidence Intervals for Daily Volatility (2/2)

- SPY, March–May 2025
  - band: 95% confidence interval for $\sqrt{252\,\mathrm{IV}_t}$ from realised quarticity
- The volatility band takes the square root of the variance endpoints
  - $\sqrt{\mathrm{RV}_t}\exp\left(\pm 0.98\,\mathrm{se}(\mathrm{RV}_t)/\mathrm{RV}_t\right)$: the log half-width is halved
- SPY: a single day's volatility is known to about $\pm 20\%$
  - median $\mathrm{se}/\mathrm{RV} = 0.19$, so the volatility factors are $[0.83, 1.20]$
  - median relative width: 37%
- 9 April 2025: volatility 72%, CI $[55\%, 93\%]$
  - the day the tariff pause was announced
  - The interval widens when a few returns dominate the sum: $\sum r^4$ grows faster than $(\sum r^2)^2$
- The band assumes a continuous price and negligible noise
  - on a jump day, $\sum r^4$ keeps the term κ^4
  - so the band has no nominal coverage for either the total variation or IV_t (section Jumps)
- RV is still far more precise than r_t^2
  - the relative error of r_t^2 is of order $\sqrt{2} \approx 141\%$ under the Normal distribution

MFM_ch9_rv_ci

Intraday Seasonality (1/2)



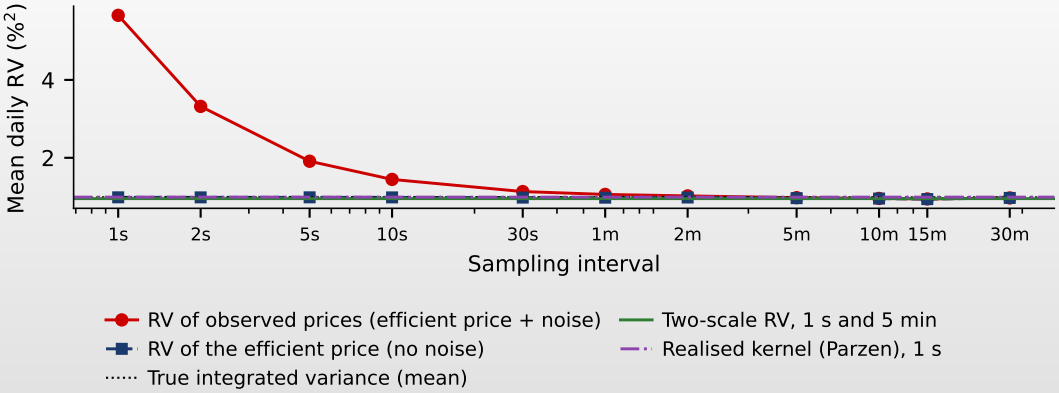
MFM_ch9_seasonality

Intraday Seasonality (2/2)

- Left: SPY, mean $|r_{t,i}|$ by 5-minute interval of the day
  - full days only
  - ticks mark the end of the interval, New York time
- Right: Bitcoin by UTC hour
- SPY: a **U-shape** Andersen & Bollerslev (1997)
  - 0.106% in the first 5 minutes, 0.050% around midday, 0.089% in the last 5 minutes
  - The opening auction incorporates overnight news; the close concentrates index and fund rebalancing
- Bitcoin trades 24/7 but is most volatile at 14:00 UTC
  - 14:00 UTC: near the US equity opening (09:30 New York: 13:30 UTC in summer, 14:30 UTC in winter)
  - quietest at 10:00 UTC (ratio 2.2)
- RV sums over the day, so seasonality does not bias it
  - it does bias bipower variation (BV, defined in section Jumps) and makes the jump test over-reject
  - the remedy, dividing returns by a periodicity factor f_i Boudt, Croux & Laurent (2011), is applied to the jump test in section Jumps (Appendix)

MFM_ch9_seasonality

The Volatility Signature Plot: Simulation (1/2)



MFM_ch9_noise_simulation

The Volatility Signature Plot: Simulation (2/2)

- A signature plot shows mean RV against the sampling interval
  - here: 100 simulated days, one price per second
  - efficient price with random daily volatility plus i.i.d. noise, $\omega = 0.01\%$
- True mean IV $= 0.98$; RV of observed prices:
  - 5.66 at 1 second, 1.06 at 1 minute, 0.98 at 5 minutes
  - The bias matches $2n\omega^2$ (slide *Why Not Use Every Tick?*)
- Without noise, RV is flat in the sampling interval
  - the rise at high frequency is the fingerprint of noise
- Noise-robust estimators on 1-second data: two-scale RV 0.95, realised kernel 0.99
  - error standard deviation across days: 5-minute RV 0.187, two-scale 0.150, kernel 0.045
  - using all data pays off only with a noise correction

MFM_ch9_noise_simulation

Subsampling and Two-Scale RV

- **Subsampling**: compute RV on K shifted grids and average
  - offsets $0, 1, \ldots, K-1$; $\mathrm{RV}_t^{(k)}$: RV on the grid with offset k
  - $\mathrm{RV}_t^{\mathsf{avg}} = \frac{1}{K} \sum_{k=0}^{K-1} \mathrm{RV}_t^{(k)}$: same bias as one sparse grid, smaller variance
- **Two-scale RV** (TSRV) Zhang, Mykland & Aït-Sahalia (2005): remove the remaining bias with the all-data RV
  - $\mathrm{TSRV}_t = \mathrm{RV}_t^{\mathsf{avg}} - \frac{\bar{n}}{n} \mathrm{RV}_t^{\mathsf{all}}$, $\bar{n} = (n - K + 1)/K$; $\mathrm{RV}_t^{\mathsf{all}}$: RV on all n returns, which estimates the noise term $2n\omega^2$
  - $\bar{n}$ is the average number of returns per sparse grid; every grid includes the first and the last price of the day
  - In small samples, divide TSRV by $1 - \bar{n}/n$

Realised Kernels



Neil Shephard, 2004

Photo: Renate Schmid (2004); CC BY-SA 2.0 de; Wikimedia Commons

- **Realised kernel** Barndorff-Nielsen, Hansen, Lunde & Shephard (2008): add weighted autocovariances of returns
  - $\mathrm{RK}_t = \gamma_0 + 2\sum_{h=1}^{H} k\left(\frac{h}{H+1}\right)\gamma_h$, $\gamma_h = \sum_i r_{t,i}\, r_{t,i-h}$
  - Parzen weight function k: 1 at 0, falling smoothly to 0 at 1; keeps RK positive
- Noise makes γ_1 negative
  - the kernel adds it back and cancels the bias
- Bandwidth H grows with the noise-to-signal ratio
  - $H \approx 3.5\,\xi^{4/5} n^{3/5}$, with $\xi^2 = \omega^2/\sqrt{\mathrm{IQ}}$ the noise-to-signal ratio

The Benjamini–Hochberg Rule on Five Tests

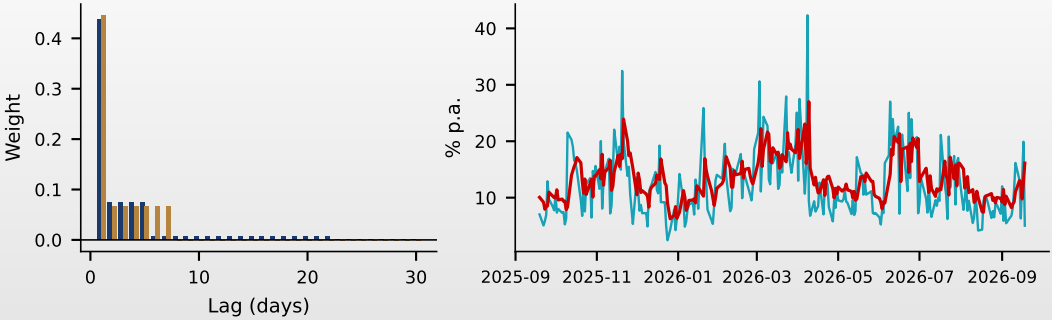
- Five days, sorted p-values $0.003, 0.012, 0.024, 0.045, 0.60$
  - with $q = 0.05$ and $N = 5$ the thresholds kq/N are $0.01, 0.02, 0.03, 0.04, 0.05$
- The first three pass, the fourth fails: $k = 3$, three days rejected
  - $0.024 \leq 0.03$, but $0.045 > 0.04$
- Same data, number of rejections
  - no correction ($p \leq 0.05$): 4
  - BH: 3
  - Bonferroni ($p \leq 0.05/5 = 0.01$): 1
- The procedure is step-up
  - with p-values $0.003, 0.012, 0.032, 0.038, 0.60$, the third fails its own threshold ($0.032 > 0.03$)
  - but the fourth passes ($0.038 \leq 0.04$), so $k = 4$ and four days are rejected

Implied Volatility as a Variance Forecast

- As a forecast: regress $\frac{252}{21}\mathrm{RV}_{t,t+21}$ on VIX_t^2
  - $\frac{252}{21}\mathrm{RV}_{t,t+21}$: total variance, annualised $\%^2$
  - VIX_t^2: annualised $\%^2$; 30 calendar days $\approx$ 21 trading days
  - slope 0.56 (standard error, SE, 0.05) < 1: VIX^2 moves more than the variance that follows
  - so VIX over-reacts, partly because the premium itself rises in turbulent periods
  - $R^2 = 0.23$: VIX is informative
  - the level is a separate question: the mean gap of 4.37 percentage points is the premium
  - the same slope-versus-level distinction returns in the Mincer–Zarnowitz test

MFM_ch9_vix

Implied Lag Weights and In-Sample Fit (1/2)



MFM_ch9_har

Implied Lag Weights and In-Sample Fit (2/2)

- Left: weights of log-HAR on each past day
  - Bitcoin: complete 7- and 30-day windows
- Right: SPY, last 250 days, realised volatility and log-HAR fitted values
- SPY, weights by lag
  - lag 1: $\hat{\beta}_d + \hat{\beta}_w/5 + \hat{\beta}_m/22 = 0.362 + 0.067 + 0.008 = 0.437$
  - each of lags 2–5: $0.067 + 0.008 = 0.075$
  - each of lags 6–22: 0.008
  - Three coefficients mimic the slow decay of the autocorrelation function
- Bitcoin: yesterday 0.446, days 2–7 about 0.066 each, almost nothing beyond a week
- In-sample fitted values follow the level of volatility, not the day-to-day noise
- HAR is an AR(22) with 19 restrictions: $\phi_2 = \ldots = \phi_5$, $\phi_6 = \ldots = \phi_{22}$
  - Newey–West Wald test on SPY: $W = 18.1$, $p = 0.51$
  - not rejected (derivation in the Appendix)

MFM_ch9_har

Extensions of HAR

- **HAR-CJ** Andersen, Bollerslev & Diebold (2007): separate continuous (C) and jump (J) parts as regressors
  - $\ln \mathrm{RV}_{t+1} = \beta_0 + \beta_{Cd} \ln C_t + \beta_{Cw} \ln C_t^{(w)} + \beta_{Cm} \ln C_t^{(m)}$
  - $\quad + \beta_{Jd} \ln(1 + J_t) + \beta_{Jw} \ln(1 + J_t^{(w)}) + \beta_{Jm} \ln(1 + J_t^{(m)}) + \varepsilon_{t+1}$
  - $C_t^{(w)}, J_t^{(w)}$ ($C_t^{(m)}, J_t^{(m)}$): arithmetic means over the last 5 (22) days, as $\mathrm{RV}_t^{(w)}$; logs are taken after averaging
- Weekend or holiday dummies
- Leverage: negative returns raise future RV
- Implied volatility as an extra regressor
- Multi-step: regress $\sum_{j=1}^{h} \mathrm{RV}_{t+j}$ directly on the same regressors

Results: Bitcoin, 2025–2026 (1/2)

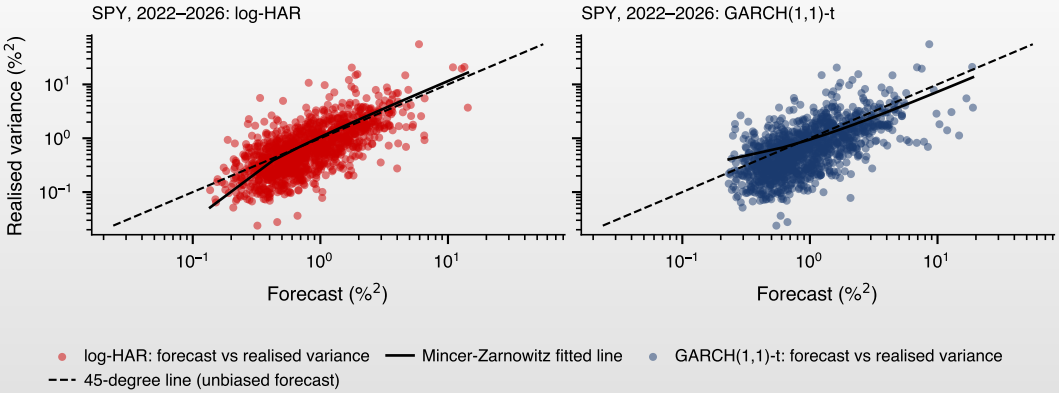
Model	QLIKE	DM_{QLIKE} t (p)	MSE ($\%^4$)	DM_{MSE} p	MZ b	MZ R^2
log-HAR	0.282	–	26.6	–	0.82	0.38
HAR	0.409	-1.22 (0.223)	27.6	0.431	0.89	0.34
GARCH(1,1)-t	0.360	-4.32 (< 0.001)	42.2	0.055	0.55	0.23
EWMA	0.346	-2.96 (0.003)	36.3	0.033	0.64	0.18
Yesterday's RV	0.466	-5.38 (< 0.001)	35.0	0.061	0.58	0.33

- 422 one-day-ahead forecasts of 24-hour realised variance
  - same models and tests as for SPY
- log-HAR again has the lowest QLIKE: 22% below GARCH(1,1)-t
  - against HAR in levels the gap is not significant ($p = 0.223$)
  - GARCH on daily data is biased: MZ $b = 0.55$ ($p < 0.001$): it overreacts to large daily moves

Results: Bitcoin, 2025–2026 (2/2)

- HAR in levels is worse than GARCH under QLIKE (0.409 vs 0.360)
  - after calm weekends it predicts too low a variance
  - QLIKE punishes under-prediction
- Under MSE only EWMA is significantly worse than log-HAR at 5% ($p = 0.033$)
  - GARCH: $p = 0.055$
- Bitcoin and SPY MSE are not comparable
  - MSE is in $(\%^2)^2 = \%^4$ and grows with the square of the variance level
  - QLIKE depends only on RV_t / F_t and is comparable across markets

Mincer–Zarnowitz Regressions (1/2)



MFM_ch9_forecasts

Mincer–Zarnowitz Regressions (2/2)

- SPY, 2022–2026: realised variance against the forecast (log scales)
  - dashed: 45-degree line
  - black: fitted regression in levels
- log-HAR forecasts span a wider range
  - they separate calm from turbulent days better
- On calm days GARCH overstates the variance
  - GARCH forecasts rarely fall below $0.26\%^2$ per day (1% of days)
  - log-HAR goes down to $0.14\%^2$
- The cloud is wide for both
  - one day's variance is hard to predict
  - RV itself is measured with error

MFM_ch9_forecasts

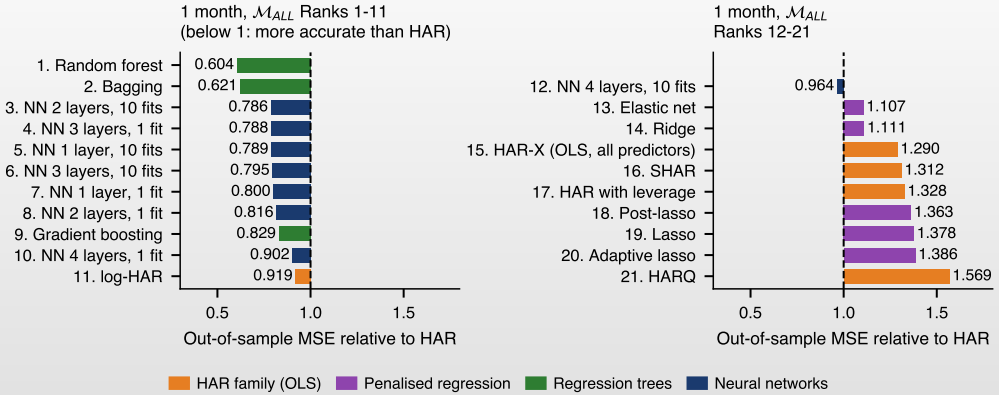
Reading the Model Confidence Set

- An MCS p-value is a running maximum
  - if a model is eliminated at step k, its MCS p-value is $\max_{j \le k} p_j$
  - p_j: the p-value of the equal-ability test at step j
  - so the MCS p-values rise along the elimination order, which the table reads off
  - equal values (GARCH, HARQ) mean the later model's own step p-value was not larger, so their relative order is not identified from the table
  - a model stays in the 90% set if its MCS p-value exceeds 0.10
- Why is HAR (QLIKE 0.326) dropped before GARCH (0.343)?
  - each step eliminates the model with the largest **standardised** loss gap to the other survivors
  - not the largest mean loss
  - SHAR differs from HAR only in its daily term, so their losses move together and the gap is very stable: DM SHAR vs HAR $t = -3.37$, $p < 0.001$
  - GARCH loses more on average, but its gaps are noisier: against log-HAR $t = -2.30$, smaller in size than the gap that separates HAR from SHAR (as in the check on the Diebold–Mariano slide)
  - the DM table compares each model only with log-HAR, so it does not show the HAR–SHAR pair, the likely reason HAR goes early

Case Study: Can Machine Learning Beat HAR?

- Models, with minimal tuning
  - the HAR family by OLS and penalised regressions
  - tree ensembles (bagging, random forest, boosting) and neural networks
  - full catalogue in the Appendix
- Evaluation: out-of-sample MSE relative to HAR
  - per stock and then averaged
  - DM tests per stock and MCS at 90%

Case Study: Ranking of 21 Models, One Month Ahead (1/2)



MFM_ch9_ml_vs_har

Case Study: Ranking of 21 Models, One Month Ahead (2/2)

- Source: Christensen, Siggaard & Veliyev (2023)
  - MSE relative to HAR, one month ahead, all 12 predictors, mean over 29 DJIA stocks
  - NN_k^e: k hidden layers, e networks averaged
- Tree ensembles lead: random forest 0.604, bagging 0.621
  - all eight networks beat HAR (0.786–0.964), but depth does not help
- Penalised regressions and linear HAR extensions lose to HAR (1.107–1.569)
- Each bar averages per-stock ratios: not a significance test of the ranking

MFM_ch9_ml_vs_har

Itô Semimartingales and Stable Convergence

- Itô semimartingale: $dp = \mu\, ds + \sigma\, dW + dJ$
  - μ, σ adapted (known from the past at each time)
  - σ càdlàg (right-continuous with left limits, so it may jump)
  - σ may depend on W (leverage)
  - Without jumps: $\sqrt{M}(\mathrm{RV}_t - \mathrm{IV}_t) \xrightarrow{\mathcal{L}\text{-}s} \mathrm{MN}(0, 2\,\mathrm{IQ}_t)$, a **mixed** Normal distribution whose variance $2\,\mathrm{IQ}_t$, with integrated quarticity IQ, $\mathrm{IQ}_t = \int \sigma^4(s)\, ds$, is itself random Barndorff-Nielsen & Shephard (2002)
- **Stable** convergence ($\mathcal{L}$-s) Mykland & Zhang (2009)
  - the limit holds jointly with every variable of the path's σ-field
  - Hence $(\mathrm{RV}_t - \mathrm{IV}_t)/\sqrt{2\,\mathrm{RQ}_t/M} \to N(0,1)$: studentising by $\sqrt{2\,\mathrm{RQ}_t/M}$, with $\mathrm{RQ}_t \to \mathrm{IQ}_t$ random, is legitimate only under stable convergence
- Finite samples: the t-statistic for $\ln \mathrm{RV}_t$ has better coverage than the one in levels Barndorff-Nielsen & Shephard (2005)
  - the confidence intervals of the lecture use it

Back

Deriving the Noise Bias and the Optimal Frequency

- Observed return $r_i = r_i^* + \varepsilon_i - \varepsilon_{i-1}$, with ε i.i.d. $(0, \omega^2)$ and independent of p^*
  - Given the path of σ (zero drift, no leverage: σ independent of W):
    $\mathbb{E}[\mathrm{RV}^{(n)} \mid \sigma] = \sum_i \mathbb{E}[r_i^{*2} \mid \sigma] + \sum_i \mathbb{E}[(\varepsilon_i - \varepsilon_{i-1})^2] = \mathrm{IV} + 2n\omega^2$
  - $\mathrm{cov}(r_i, r_{i-1}) = -\mathbb{E}[\varepsilon_{i-1}^2] = -\omega^2$
  - By a law of large numbers for the noise terms, $\mathrm{RV}^{(n)}/(2n) \xrightarrow{p} \omega^2$ as $n \to \infty$: the all-tick RV estimates the noise, not IV
    Bandi & Russell (2008)
- MSE: $\mathbb{E}[(\mathrm{RV}^{(n)} - \mathrm{IV})^2] \approx \underbrace{2\,\mathrm{IQ}/n}_{\text{discretisation}} + \underbrace{4n^2\omega^4}_{\text{squared bias}} + O(n)$
  - First-order condition $-2\,\mathrm{IQ}/n^2 + 8n\omega^4 = 0$ gives $n^* = (\mathrm{IQ}/(4\omega^4))^{1/3}$ Bandi & Russell (2008)

Back

Intraday Seasonality and the Jump Test

- Seasonality biases BV (not RV)
  - $\mathbb{E}|r_i|\,|r_{i-1}| \propto \sigma_i \sigma_{i-1} \leq (\sigma_i^2 + \sigma_{i-1}^2)/2$
  - Fast changes of scale (open, close) push $\mathrm{RV} - \mathrm{BV}$ up, so the ratio test (section Jumps) over-rejects
- Periodicity factor f_i: standard-deviation estimator of Boudt, Croux & Laurent (2011)
  - scale each return by $\sqrt{\mathrm{BV}_t / M_t}$
  - take the root mean square by interval of the day
  - normalise the mean of f_i^2 to 1
  - SPY: $f = 1.78$ at the open, 0.73 at the minimum, 1.52 at the close
- Dividing $r_{t,i}$ by f_i before the test, at the 0.1% level 20 SPY rejections remain
  - for the 5% level: main text

Back

Rates of Convergence under Noise

Estimator	Error of order	Positive	Source
RV on all ticks	inconsistent (bias $2n\omega^2$)	yes	Bandi & Russell (2008)
Sparse RV, fixed 5-minute grid	$M^{-1/2}$, M fixed; bias stays	yes	Liu, Patton & Sheppard (2015)
Two-scale RV	$n^{-1/6}$	no	Zhang, Mykland & Aït-Sahalia (2005)
Multi-scale RV	$n^{-1/4}$	no	Zhang (2006)
Realised kernel, flat-top / Parzen	$n^{-1/4}$ / $n^{-1/5}$	no / yes	Barndorff-Nielsen, Hansen, Lunde & Shephard (2008)
Pre-averaging	$n^{-1/4}$	no	Jacod, Li, Mykland, Podolskij & Vetter (2009)

- $n^{-1/4}$ is the optimal rate with i.i.d. noise
  - the price of noise is half the usual exponent
- With $n = 23\,400$ seconds: $n^{1/4} \approx 12.4$, against $\sqrt{78} \approx 8.8$ for 5-minute RV
  - a heuristic only: rates ignore constants, noise size and the bias of sparse RV
  - consistent with the modest gains in Liu, Patton & Sheppard (2015)

Back

Pre-Averaging (1/2)

- Local averages of $k_n \approx \theta\sqrt{n}$ returns with a weight g on $[0,1]$ Jacod, Li, Mykland, Podolskij & Vetter (2009)
  - θ: a tuning constant, not the jump-test θ
  - e.g. $g(x) = \min(x, 1-x)$
  - $\bar{r}_i = \sum_{j=1}^{k_n - 1} g(j/k_n)\, r_{i+j}$: noise variance shrinks like $1/k_n$, the signal keeps order k_n/n
- Estimator, with $\psi_1 = \int_0^1 g'(x)^2 dx$ and $\psi_2 = \int_0^1 g(x)^2 dx$
  - $\psi_1 = 1$ and $\psi_2 = 1/12$ for the triangular g; g': the derivative of g
  - $\widehat{\mathrm{IV}} = \frac{1}{\theta\, \psi_2 \sqrt{n}} \sum_i \bar{r}_i^2 - \frac{\psi_1}{2\,\theta^2 \psi_2\, n} \sum_i r_i^2$ (second term: noise bias correction)

Back

Pre-Averaging (2/2)

- Rate $n^{-1/4}$
  - the same device gives jump-robust (pre-averaged bipower) and covariance estimators
- Built for tick data
  - on 5-minute bars little noise is left to average out (empirical signature plot, section Microstructure Noise)

Back

Realised Covariance and Asynchronous Trading

- Realised covariance $\sum_i r^{(1)}_{t,i} r^{(2)}_{t,i} \to \int \sigma_{12}(s)\, ds$ needs prices of both assets on the **same** grid
  - $r^{(1)}, r^{(2)}$: the returns of assets 1 and 2, $\sigma_{12}(s)$: their spot covariance
  - Epps effect Epps (1979): measured correlation falls towards 0 as sampling gets finer, because trades are asynchronous and the last price is stale
- Hayashi & Yoshida (2005): sum the products of **all** pairs of returns whose time intervals overlap
  - $\sum_{i,j} r^{(1)}_i r^{(2)}_j 1\{(t_{i-1}, t_i] \cap (s_{j-1}, s_j] \neq \emptyset\}$: $(t_{i-1}, t_i]$, $(s_{j-1}, s_j]$: the intervals between consecutive trades of each asset
  - unbiased without synchronisation (in the absence of noise)
- With noise as well: pre-averaged or multivariate kernel versions
  - application: intraday portfolio risk (Chapter 4)

Back

Locating Jumps Within the Day, Lee–Mykland (1/2)

- Lee & Mykland (2008): $L_{t,i} = r_{t,i}/\hat{\sigma}_{t,i}$
  - $\hat{\sigma}^2_{t,i}$: the bipower average of the previous $K = 270$ returns (their value for 5-minute data)
  - Under no jumps, $\max_i |L_i|$ over n returns has a Gumbel limit: reject if $(|L| - C_n)/S_n > -\ln(-\ln(1-\alpha))$
  - $C_n = \sqrt{2\ln n} - \frac{\ln\pi + \ln\ln n}{2\sqrt{2\ln n}}$ and $S_n = 1/\sqrt{2\ln n}$: the centring and scaling constants of the Gumbel limit
  - our $\hat{\sigma}^2$ includes the factor $\pi/2$, so the constant $c = \sqrt{2/\pi}$ of Lee & Mykland (2008) becomes 1
  - SPY, returns divided by f_i, $\alpha = 1\%$, $n = 115\,453$ (the whole sample): threshold $|L| > 5.41$

Back

Locating Jumps Within the Day, Lee–Mykland (2/2)

- 169 jumps on 105 days, 62 of them on Wednesdays between 14:00 and 15:00
  - when Federal Open Market Committee (FOMC) statements (14:00) and press conferences (14:30) are released
  - Largest: $L = -22.9$ on 2025-10-29
  - only 10% of these days are also flagged by the daily ratio test
- The two tests ask different questions
  - Lee–Mykland: one extreme return relative to local volatility
  - ratio test: a daily gap between RV and BV

Back

Jump-Robust Estimators of Integrated Variance (1/2)

- Truncated RV Mancini (2009): $\text{TRV}_t = \sum_i r_{t,i}^2 \, 1\{|r_{t,i}| \leq c \, \hat{\sigma}_t f_i \, \Delta^{\varpi}\}$, $\varpi \in (0, 1/2)$
  - Keep only the squared returns below a threshold: c: a multiple, $\hat{\sigma}_t$: a jump-robust daily volatility, f_i: the periodicity factor, ϖ: the exponent of the interval length Δ
  - Diffusive increments are $O_p(\Delta^{1/2})$ (of order $\sqrt{\Delta}$) and end up below the threshold; jumps are $O(1)$ and are cut off

Back

Jump-Robust Estimators of Integrated Variance (2/2)

- MedRV Andersen, Dobrev & Schaumburg (2012): $\frac{\pi}{6-4\sqrt{3}+\pi} \frac{M}{M-2} \sum_i \mathrm{med}(|r_{i-1}|, |r_i|, |r_{i+1}|)^2$
  - med: the median of three adjacent absolute returns; the first fraction makes MedRV consistent for IV
  - An isolated return larger than both neighbours never enters the median (which shifts to a neighbour), so a large isolated jump is excluded, though its effect is not removed exactly in finite samples
  - MedRV is less sensitive than BV to isolated zero returns
  - two adjacent jumps can still enter the median
- Jump share of SPY RV, summed over all days without a test
  - jump part: RV minus the robust estimator
  - BV 6.3%, MedRV 7.2%, TRV 12.5%
  - significant jump days only: 2.8%
- TRV with $c = 3$, $\varpi = 0.49$: truncation bites on 44% of days
  - c is a tuning choice
- Aït-Sahalia & Jacod (2009): compare power variations $\sum |r|^p$, $p > 2$, at Δ and $k\Delta$
  - the ratio tends to 1 with jumps and to $k^{p/2-1}$ without
  - known limits under both hypotheses

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HAR as a Restricted AR(22), the Wald Test

- Unrestricted: $\ln \mathrm{RV}_{t+1} = \phi_0 + \sum_{k=1}^{22} \phi_k \ln \mathrm{RV}_{t+1-k} + \varepsilon_{t+1}$
  - HAR imposes $\phi_2 = \ldots = \phi_5$ (3 restrictions) and $\phi_6 = \ldots = \phi_{22}$ (16): $R\phi = 0$ with 19 rows
- Wald: $W = (R\hat{\phi})^\top (R\hat{V}_{\mathrm{NW}} R^\top)^{-1} R\hat{\phi} \to \chi^2_{19}$ under the HAR restrictions
  - R: the 19×23 restriction matrix, $\hat{V}_{\mathrm{NW}}$: Newey–West covariance of $\hat{\phi}$
  - large W = restrictions rejected
- SPY log total variance: $W = 18.1$, $p = 0.51$
  - R^2 0.477 (AR(22)) vs 0.470 (HAR)
  - The step restrictions are not rejected
  - HAR saves 19 parameters at almost no in-sample cost
  - less estimation noise helps out of sample

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Measurement Error in the Regressors and HARQ (1/2)

- $\mathrm{RV}_t = \mathrm{IV}_t + u_t$, with $\mathbb{E}[u_t \mid \sigma] = 0$ and $\mathrm{Var}(u_t \mid \sigma) = 2\,\mathrm{IQ}_t/M$
  - the CLT of the section Realised Variance
  - If $\mathrm{IV}_{t+\mathbf{1}} = \beta_{\mathbf{0}} + \beta\,\mathrm{IV}_t + \eta_{t+\mathbf{1}}$, OLS of $\mathrm{RV}_{t+\mathbf{1}}$ on RV_t gives plim $\hat{\beta} = \beta\,\dfrac{\mathrm{Var}(\mathrm{IV})}{\mathrm{Var}(\mathrm{IV}) + \mathbb{E}[2\,\mathrm{IQ}/M]} < \beta$ for $\beta > 0$: attenuation ($\eta_{t+\mathbf{1}}$: the shock to IV; plim: probability limit)
- If the signal variance does not rise with IQ_t, the signal-to-noise ratio of RV_t falls when IQ_t is high
  - a working approximation
  - HARQ approximates the falling weight by $\beta_d + \beta_Q\sqrt{\mathrm{RQ}_t}$, $\beta_Q < 0$ Bollerslev, Patton & Quaedvlieg (2016)

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Measurement Error in the Regressors and HARQ (2/2)

- SPY, levels, Newey–West SE
  - intraday RV: $\hat{\beta}_Q = -0.0088$ (0.0042)
  - total variance: $\hat{\beta}_Q = -0.0022$ (0.0024)
  - Daily slope (intraday RV) 1.13 at the median $\sqrt{\text{RQ}}$, 1.09 at its 99th percentile: the sign predicted by the theory, a small effect
- Read $\hat{\beta}_d$ together with the other slopes
  - the daily slope above 1 (1.135) is partly offset by a negative weekly slope (-0.120)
  - the three RV averages overlap by construction (each contains yesterday's RV), so the split is imprecise (SE of $\hat{\beta}_d$: 0.494)
  - the slopes sum to 1.04 (total variance: 0.80), so the intraday levels fit is an in-sample diagnostic for the sign of β_Q, not a forecasting model
- Like for like, total variance: HAR $\hat{\beta}_d = 0.399$, HARQ $\hat{\beta}_d = 0.542$
  - letting the weight fall with RQ raises the base daily slope, in the direction the attenuation argument predicts
  - the difference is within one SE

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Beyond HAR, Semivariances and Joint Return–RV Models (1/2)

- Realised semivariances enter SHAR Patton & Sheppard (2015): $\beta_d \mathrm{RV}_t$ becomes $\beta^+ \mathrm{RS}_t^+ + \beta^- \mathrm{RS}_t^-$
  - $\mathrm{RS}_t^{\pm} = \sum_i r_{t,i}^2 \, 1\{r_{t,i} \gtrless 0\}$, $\mathrm{RS}^+ + \mathrm{RS}^- = \mathrm{RV}$
  - SPY, levels (overnight return counted as one more return of the day): $\hat{\beta}^- = 0.56$ (0.23), $\hat{\beta}^+ = 0.21$ (0.18)
  - Wald $\beta^+ = \beta^-$: $p = 0.35$
  - Bad volatility is more persistent, as in Patton & Sheppard (2015), but six years of data do not separate the two slopes
  - In the MCS comparison, HARQ and SHAR are estimated in levels with the filter of Bollerslev, Patton & Quaedvlieg (2016): forecasts outside the [min, max] range of the estimation-sample values are replaced by the sample mean

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Beyond HAR, Semivariances and Joint Return–RV Models (2/2)

- Realized GARCH Hansen, Huang & Shek (2012): a GARCH equation plus a **measurement equation**
  - the GARCH equation is driven by the realised measure x_t
  - $\ln h_t = \omega + \beta \ln h_{t-\mathbf{1}} + \gamma \ln x_{t-\mathbf{1}}$, $\ln x_t = \xi + \varphi \ln h_t + \tau(z_t) + u_t$, $r_t = \sqrt{h_t}\, z_t$: joint quasi-maximum likelihood
  - h_t: conditional variance; x_t: realised measure (e.g. RV); z_t: standardised shock; $\tau(z_t)$: leverage function
  - ξ, φ: measurement-equation coefficients, u_t: its error
- HEAVY (high-frequency-based volatility) models Shephard & Sheppard (2010): consistent multi-step forecasts
  - one GARCH-type equation for the daily variance, driven by lagged RV
  - one equation for RV itself

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Case Study Data, 29 DJIA Stocks and 12 Predictors

- Data: 29 of the 30 DJIA stocks (Visa excluded), 29 January 2001 – 31 December 2017
  - $T = 4\,257$ days
  - 5-minute RV from cleaned New York Stock Exchange (NYSE) trade prices
  - $n = 78$ returns a day
- Split: training 70% (2 964 days), validation 10% (424), test 20% (847)
  - $2\,964 + 424 + 847 = 4\,235$: the first 22 days only feed the monthly RV average
  - robustness: training 1 000 or 2 000 days, validation 200
- $\mathcal{M}_{\mathbf{HAR}}$: daily, weekly, monthly RV
- $\mathcal{M}_{\mathbf{ALL}}$ adds 9 predictors
  - implied volatility, earnings-day dummy, 1-week momentum, change in log dollar volume
  - VIX, squared Hang Seng return, business-conditions index, change in the 3-month T-bill rate, policy-uncertainty index

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Case Study Models, from HAR to Neural Networks

- Models: HAR, log-HAR, HAR with leverage, SHAR, HARQ
  - by OLS on a rolling window
- Penalised: ridge, lasso, elastic net, adaptive and post-lasso
  - ridge: L_2 penalty; lasso: L_1; elastic net: a mix
  - penalties tuned on the validation set
- Trees: bagging, random forest, gradient boosting
  - random forest with Breiman's defaults: 500 trees, $J/3$ candidate predictors per split (J: number of predictors), minimum node size 5
- Neural networks NN_k^e: $k = 1, \ldots, 4$ hidden layers (2; 4-2; 8-4-2; 16-8-4-2 neurons)
  - leaky ReLU, Adam, dropout, early stopping
  - $e = 1$: best of 100 seeds on validation, $e = 10$: average of the best 10
- Evaluation: out-of-sample MSE relative to HAR
  - averaged over stocks
  - one-sided DM test per stock Diebold & Mariano (1995)
  - MCS at 90% Hansen, Lunde & Nason (2011)

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Case Study, Further Results

- One day ahead, RV lags only
  - MSE relative to HAR: log-HAR 0.995, HARQ 1.059, random forest 1.020, neural networks 0.954–0.990
  - Past RV leaves little to learn
  - the networks give the clearest gains (1–5%), log-HAR barely beats HAR
- One day ahead, all 12 predictors: elastic net 0.916
  - the other models are on the horizon chart of the case study
  - Extra predictors pay most with shrinkage, nonlinearity or the log transform
  - HARQ overfits
- MCS at 90%, with all predictors
  - the networks stay in the set for about 80% of stocks
  - HAR stays for about 20–40% of stocks
  - Variable importance from accumulated local effects: daily RV, weekly RV and implied volatility dominate in all four models shown
  - Random forest and networks rank implied volatility higher than OLS with all predictors and elastic net

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Case Study, Replication, Limits and Extensions (1/2)

- Replicate on course data: SPY 5-minute bars
  - open-to-close RV with 78 returns a day
  - the authors report that SPY confirms the stock results
  - The authors' robustness split: 1 000 training days, 200 validation days, the rest to 18 September 2026 as test set
  - Predictors: VIX (computed from S&P 500 index options; a proxy for SPY's own implied volatility, which would need SPY options), squared Hang Seng return, SPY momentum and dollar volume, T-bill and policy uncertainty (Federal Reserve Economic Data, FRED)
  - no earnings dummy for a fund

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Case Study, Replication, Limits and Extensions (2/2)

- Limits stated by the authors
  - Minimal tuning: more tuning could improve ML further
  - the networks are trained once and not re-estimated in the test window
  - Rankings depend on the split: with 1 000 training days the financial crisis falls in the test set and the HAR extensions beat HAR
  - HAR models with a separate jump component are not included
  - accumulated local effects describe the average prediction and miss rare events such as earnings days
- Open extension: the paper ranks variance forecasts by MSE only
  - does the ML advantage survive QLIKE, which is robust to the noisy RV proxy Patton (2011)?
  - does it hold for Bitcoin's 24-hour RV?

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The Discovery Loop, Reusable Prompts (1/2)

- Tools: an AI assistant
  - a large language model (LLM), such as Claude, ChatGPT, Gemini or GitHub Copilot
  - literature search: Semantic Scholar, Elicit, Google Scholar
- **1. Literature**: tabulate the evidence
  - then check every DOI on Crossref (the title must match)
  - `Tabulate the estimates of H published since 2018: estimator, sampling frequency, treatment of the RV measurement error, point estimate and standard error, DOI.`
- **2. Competing hypotheses**, each with its implied shape of $m(2, \Delta)$
  - `Derive m(2, Delta) for three processes that give H near 0.1 with the scaling estimator: fractional log-volatility with H = 0.1; H = 0.5 plus i.i.d. error; a two-factor model with one fast mean-reverting factor.`
- **3. Data and code**: replicate the chapter's number first
  - `Write Python that builds daily total variance from the SPY 5-minute bars of the course data (5-minute RV plus the squared overnight return, previous close to today's open), computes m(q, Delta) for q in {0.5, 1, 1.5, 2, 3}, Delta = 1..30, and the slope H.` Target: $\hat{H} = 0.11$

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The Discovery Loop, Reusable Prompts (2/2)

- **4. Robustness**, pre-registered, with a power check
  - `Write a pre-registered plan: lag windows [1,30], [5,30], [10,50]; RV, bipower variation and 30-minute subsampled RV; a fit m(2, Delta) = c Delta^(2H) + k; block-bootstrap intervals; size and power by simulation at H = 0.1 and 0.5.`
- **5. Critique**: ask for the strongest counter-argument
  - `Act as a referee: which non-i.i.d. error structures in log RV (serially correlated, level-dependent, driven by overnight returns or jumps) would survive the constant-k correction, and how would you detect each?`
- **6. Report**: all windows and measures, bootstrap intervals, the simulation design, code and the AI log

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Further Reading

- Textbook and surveys
  - Franke, Härdle & Hafner (2019): statistics of financial markets, volatility models
  - Andersen, Bollerslev, Diebold & Labys (2003): modelling and forecasting realised volatility
  - Corsi (2009): the HAR model
- Crypto-assets
  - Intraday information can enter Bitcoin risk forecasts; the entropy of high-frequency returns has been proposed as one such signal Pele & Mazurencu-Marinescu-Pele (2019)
  - Crypto-assets form a distinct asset class by the features of their return distributions Pele et al. (2023)

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Further Study on Quantinar

Advanced courses on quantinar.com

- Nonparametric and Semiparametric Models
- Kalman Filter
- Applied Time Series Analysis with Python

Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_09

Acronyms used in this material (1/2)

- **ACF**: AutoCorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **BV**: Bipower Variation
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CI**: Confidence Interval
- **CLT**: Central Limit Theorem
- **DJIA**: Dow Jones Industrial Average
- **DM**: Diebold–Mariano (test)
- **DOI**: Digital Object Identifier
- **ETF**: Exchange-Traded Fund
- **EWMA**: Exponentially Weighted Moving Average
- **FDR**: False Discovery Rate
- **FOMC**: Federal Open Market Committee
- **FRED**: Federal Reserve Economic Data
- **FWER**: Family-Wise Error Rate
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- **GMM**: Generalized Method of Moments
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR**: Heterogeneous AutoRegressive (model)
- **HAR-CJ**: HAR model with Continuous and Jump components
- **HAR-RV**: Heterogeneous AutoRegressive model of Realised Volatility
- **HAR-X**: Heterogeneous AutoRegressive model with additional regressors
- **HARQ**: HAR model with Realised Quarticity

Acronyms used in this material (2/2)

- **HEAVY**: High-frEquency-bAsed VolatilitY (models)
- **IQ**: Integrated Quarticity
- **LLM**: Large Language Model
- **MAE**: Mean Absolute Error
- **MCS**: Model Confidence Set
- **ML**: Machine Learning
- **MSE**: Mean Squared Error
- **MZ**: Mincer–Zarnowitz (regression)
- **NYSE**: New York Stock Exchange
- **OLS**: Ordinary Least Squares
- **QLIKE**: Quasi-LIKElihood (loss function)
- **RK**: Realised Kernel
- **RQ**: Realised Quarticity
- **RS**: Realised Semivariance
- **RV**: Realised Variance
- **SE**: Standard Error
- **SHAR**: Semivariance HAR model
- **TQ**: Tripower Quarticity
- **TRV**: Truncated Realised Variance
- **TSRV**: Two-Scale Realised Variance
- **US**: United States
- **UTC**: Coordinated Universal Time (Temps Universel Coordonné)
- **VIX**: CBOE Volatility IndeX
- **VRP**: Variance Risk Premium

References (1/6)

- Aït-Sahalia, Y., Jacod, J. (2009). Testing for jumps in a discretely observed process. *Annals of Statistics*, 37(1), 184–222.
- Andersen, T.G., Bollerslev, T. (1997). Intraday periodicity and volatility persistence in financial markets. *Journal of Empirical Finance*, 4(2–3), 115–158.
- Andersen, T.G., Bollerslev, T. (1998). Answering the skeptics: yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 39(4), 885–905.
- Andersen, T.G., Bollerslev, T., Diebold, F.X. (2007). Roughing it up: including jump components in the measurement, modeling, and forecasting of return volatility. *Review of Economics and Statistics*, 89(4), 701–720.
- Andersen, T.G., Bollerslev, T., Diebold, F.X., Ebens, H. (2001). The distribution of realized stock return volatility. *Journal of Financial Economics*, 61(1), 43–76.
- Andersen, T.G., Bollerslev, T., Diebold, F.X., Labys, P. (2001). The distribution of realized exchange rate volatility. *Journal of the American Statistical Association*, 96(453), 42–55.
- Andersen, T.G., Bollerslev, T., Diebold, F.X., Labys, P. (2003). Modeling and forecasting realized volatility. *Econometrica*, 71(2), 579–625.
- Andersen, T.G., Dobrev, D., Schaumburg, E. (2012). Jump-robust volatility estimation using nearest neighbor truncation. *Journal of Econometrics*, 169(1), 75–93.
- Bajgrowicz, P., Scaillet, O., Treccani, A. (2016). Jumps in high-frequency data: spurious detections, dynamics, and news. *Management Science*, 62(8), 2198–2217.
- Bandi, F.M., Russell, J.R. (2008). Microstructure noise, realized variance, and optimal sampling. *Review of Economic Studies*, 75(2), 339–369.

References (2/6)

- Barndorff-Nielsen, O.E., Hansen, P.R., Lunde, A., Shephard, N. (2008). Designing realized kernels to measure the ex post variation of equity prices in the presence of noise. *Econometrica*, 76(6), 1481–1536.
- Barndorff-Nielsen, O.E., Shephard, N. (2002). Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society, Series B*, 64(2), 253–280.
- Barndorff-Nielsen, O.E., Shephard, N. (2004). Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics*, 2(1), 1–37.
- Barndorff-Nielsen, O.E., Shephard, N. (2005). How accurate is the asymptotic approximation to the distribution of realised variance? In D.W.K. Andrews, J.H. Stock (Eds.), *Identification and Inference for Econometric Models*, Cambridge University Press, 306–331.
- Barndorff-Nielsen, O.E., Shephard, N. (2006). Econometrics of testing for jumps in financial economics using bipower variation. *Journal of Financial Econometrics*, 4(1), 1–30.
- Benjamini, Y., Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society, Series B*, 57(1), 289–300.
- Benjamini, Y., Yekutieli, D. (2001). The control of the false discovery rate in multiple testing under dependency. *The Annals of Statistics*, 29(4), 1165–1188.
- Bennedsen, M., Lunde, A., Pakkanen, M.S. (2022). Decoupling the short- and long-term behavior of stochastic volatility. *Journal of Financial Econometrics*, 20(5), 961–1006.
- Bolko, A.E., Christensen, K., Pakkanen, M.S., Veliyev, B. (2023). A GMM approach to estimate the roughness of stochastic volatility. *Journal of Econometrics*, 235(2), 745–778.
- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.

References (3/6)

- Bollerslev, T. (1987). A conditionally heteroskedastic time series model for speculative prices and rates of return. *Review of Economics and Statistics*, 69(3), 542–547.
- Bollerslev, T., Patton, A.J., Quaedvlieg, R. (2016). Exploiting the errors: a simple approach for improved volatility forecasting. *Journal of Econometrics*, 192(1), 1–18.
- Bollerslev, T., Tauchen, G., Zhou, H. (2009). Expected stock returns and variance risk premia. *Review of Financial Studies*, 22(11), 4463–4492.
- Boudt, K., Croux, C., Laurent, S. (2011). Robust estimation of intraweek periodicity in volatility and jump detection. *Journal of Empirical Finance*, 18(2), 353–367.
- Carr, P., Wu, L. (2009). Variance risk premiums. *Review of Financial Studies*, 22(3), 1311–1341.
- Christensen, K., Oomen, R.C.A., Podolskij, M. (2014). Fact or friction: jumps at ultra high frequency. *Journal of Financial Economics*, 114(3), 576–599.
- Christensen, K., Siggaard, M., Veliyev, B. (2023). A machine learning approach to volatility forecasting. *Journal of Financial Econometrics*, 21(5), 1680–1727.
- Cont, R., Das, P. (2024). Rough volatility: fact or artefact? *Sankhya B*, 86(1), 191–223.
- Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2), 174–196.
- Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.

References (4/6)

- Duan, N. (1983). Smearing estimate: a nonparametric retransformation method. *Journal of the American Statistical Association*, 78(383), 605–610.
- Epps, T.W. (1979). Comovements in stock prices in the very short run. *Journal of the American Statistical Association*, 74(366), 291–298.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). Statistics of Financial Markets: An Introduction, 5th ed. Springer.
- Fukasawa, M., Takabatake, T., Westphal, R. (2022). Consistent estimation for fractional stochastic volatility model under high-frequency asymptotics. *Mathematical Finance*, 32(4), 1086–1132.
- Gatheral, J., Jaisson, T., Rosenbaum, M. (2018). Volatility is rough. *Quantitative Finance*, 18(6), 933–949.
- Hansen, P.R., Huang, Z., Shek, H.H. (2012). Realized GARCH: a joint model for returns and realized measures of volatility. *Journal of Applied Econometrics*, 27(6), 877–906.
- Hansen, P.R., Lunde, A. (2005). A forecast comparison of volatility models: does anything beat a GARCH(1,1)? *Journal of Applied Econometrics*, 20(7), 873–889.
- Hansen, P.R., Lunde, A. (2006). Realized variance and market microstructure noise. *Journal of Business & Economic Statistics*, 24(2), 127–161.
- Hansen, P.R., Lunde, A., Nason, J.M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497.
- Hayashi, T., Yoshida, N. (2005). On covariance estimation of non-synchronously observed diffusion processes. *Bernoulli*, 11(2), 359–379.

References (5/6)

- Huang, X., Tauchen, G. (2005). The relative contribution of jumps to total price variance. *Journal of Financial Econometrics*, 3(4), 456–499.
- Jacod, J., Li, Y., Mykland, P.A., Podolskij, M., Vetter, M. (2009). Microstructure noise in the continuous case: the pre-averaging approach. *Stochastic Processes and their Applications*, 119(7), 2249–2276.
- Lee, S.S., Mykland, P.A. (2008). Jumps in financial markets: a new nonparametric test and jump dynamics. *Review of Financial Studies*, 21(6), 2535–2563.
- Liu, L.Y., Patton, A.J., Sheppard, K. (2015). Does anything beat 5-minute RV? A comparison of realized measures across multiple asset classes. *Journal of Econometrics*, 187(1), 293–311.
- Mancini, C. (2009). Non-parametric threshold estimation for models with stochastic diffusion coefficient and jumps. *Scandinavian Journal of Statistics*, 36(2), 270–296.
- Merton, R.C. (1980). On estimating the expected return on the market: an exploratory investigation. *Journal of Financial Economics*, 8(4), 323–361.
- Mincer, J., Zarnowitz, V. (1969). The evaluation of economic forecasts. In J. Mincer (Ed.), *Economic Forecasts and Expectations*, NBER, 3–46.
- Müller, U.A., Dacorogna, M.M., Davé, R.D., Olsen, R.B., Pictet, O.V., von Weizsäcker, J.E. (1997). Volatilities of different time resolutions: analyzing the dynamics of market components. *Journal of Empirical Finance*, 4(2–3), 213–239.
- Mykland, P.A., Zhang, L. (2009). Inference for continuous semimartingales observed at high frequency. *Econometrica*, 77(5), 1403–1445.
- Patton, A.J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256.

References (6/6)

- Patton, A.J., Sheppard, K. (2015). Good volatility, bad volatility: signed jumps and the persistence of volatility. *Review of Economics and Statistics*, 97(3), 683–697.
- Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019). Using high-frequency entropy to forecast Bitcoin's daily value at risk. *Entropy*, 21(2), 102.
- Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets? *European Journal of Finance*, 29(10), 1064–1105.
- Shephard, N., Sheppard, K. (2010). Realising the future: forecasting with high-frequency-based volatility (HEAVY) models. *Journal of Applied Econometrics*, 25(2), 197–231.
- Zhang, L. (2006). Efficient estimation of stochastic volatility using noisy observations: a multi-scale approach. *Bernoulli*, 12(6), 1019–1043.
- Zhang, L., Mykland, P.A., Aït-Sahalia, Y. (2005). A tale of two time scales: determining integrated volatility with noisy high-frequency data. *Journal of the American Statistical Association*, 100(472), 1394–1411.